AIROC™ CYW55913 hardware design guidelines

About this document

Scope and purpose

This document provides PCB layout guidelines for a board design based on AIROC™ CYW55913/55912/55911/55903/55902/55901 Wi-Fi & Bluetooth® LE Connected MCU based on the reference schematic and layout file [2].

Note: *This is a preliminary version; a later revision may include additional updates.*

Intended audience

This document is intended for hardware engineers creating PCB layout designs with CYW55913 family of chipsets.

Table of contents

About this document **1**
Table of contents **2**
1 Introduction **3**
1.1 Acronyms and abbreviations 3
1.2 IoT resources 3
1.3 Device description 3
1.4 CYW55913 WLBGA package ball map 5
1.5 Power topology 6
2 Reference design board **7**
2.1 Block diagram 7
2.2 REG_ON Delay circuit requirement 8
2.3 Major components in the reference design 8
2.4 Board layers 8
2.4.1 Layer 1 (component placement) 9
2.4.2 Layer 2 (ground plane) 11
2.4.3 Layer 3 (power plane) 12
2.4.4 Layer 4 (signal/ground plane) 13
2.4.5 Layer 5 (signal plane) 14
2.4.6 Layer 6 (signal plane) 15
3 Component selection **16**
3.1 External components selection 16
3.2 ABUCK inductance 16
3.2.1 Inductor comparative criteria 18
3.3 External bypass capacitance 19
4 Routing guidelines **20**
4.1 General PCB design guidelines 20
4.2 Routing the VBAT and ABUCK networks 20
4.3 Routing the VDDOUT_RF3P3 network 23
4.4 Routing the VDDOUT_PA3P3 network 24
4.5 Routing the ABUCK_1P12 network 27
4.6 Routing the VDDOUT_BT3P3 trace 29
4.7 Routing the VDDOUT_CLDO traces 31
4.8 Routing the crystal traces 33
4.9 Routing the RF traces 34
4.10 RF front end area 35
4.11 Strapping and Routing of the SDIO interfaces 35
4.12 Routing the SMIF signals 37
References **41**
Revision history **42**
Disclaimer **43**

1 Introduction

This guide contains examples, guidelines, and requirements to help board designers to:

- Facilitate optimal board routing and chip performance
- Address noise coupling/electromagnetic interference (EMI) current-flow capability and power management unit (PMU) performance
- Identify the comparative differences of various components (footprint and rating tradeoff) and the impact on the overall PMU functionality and performance

1.1 Acronyms and abbreviations

In most cases, acronyms and abbreviations are defined on first use. For a comprehensive list of acronyms and other terms, go to the Glossary.

1.2 IoT resources

The wealth of data available here will help you to select the right IoT device for your design, and quickly and effectively integrate the device into your design. You have access to a wide range of information, including technical documentation, schematic diagrams, product bill of materials, PCB layout information, and software updates. You can acquire technical documentation and software from the Support Community website.

1.3 Device description

This section contains a subset of the CYW55913/55912/55911/55903/55902/55901 - AIROC™ Wi-Fi & Bluetooth® LE Connected MCU chip information. For more information, see the AIROC™ Wi-Fi & Bluetooth® LE Connected MCU Low-power, 1 x 1, tri-band, IEEE 802.11ax, Wi-Fi 6/6E, Bluetooth® Low Energy 5.4 datasheet [1].

The Infineon CYW55913/55912/55911/55903/55902/55901 are a family of ultra-lower power, single-chip, connected MCUs that support 1x1 Wi-Fi 6/6E, Bluetooth® Low Energy 5.4, Matter, IP networking, with integrated PMU, targeted at Internet-of-Things (IoT) applications for stand-alone operation or to offload a host-processor. An integrated 192 MHz Arm® Cortex®-CM33, runs the Wi-Fi and Networking Stacks, Bluetooth® LE 5.3 and supports a wide array of peripherals.

A 4-wire UART or SDIO (shared with Wi-Fi) interface is available for interfacing with the host processor. All package options of the CYW55913/55912/55911/55903/55902/55901 chip provide support for SDIO v3.0, including the new UHS-I modes.

The CYW55913/55912/55911/55903/55902/55901 architecture block diagram in Figure 1 shows a simplified view of the interconnection between subsystems and blocks. CYW55913/55912/55911/55903/55902/55901 has two major connectivity subsystems: the connectivity processor + or, the Bluetooth® LE subsystem, and the Wi-Fi subsystem. The other major subsystems are I/O subsystem, peripherals, and I/O. The color-coding shows the lowest-power mode where the particular block is still functional.

Introduction

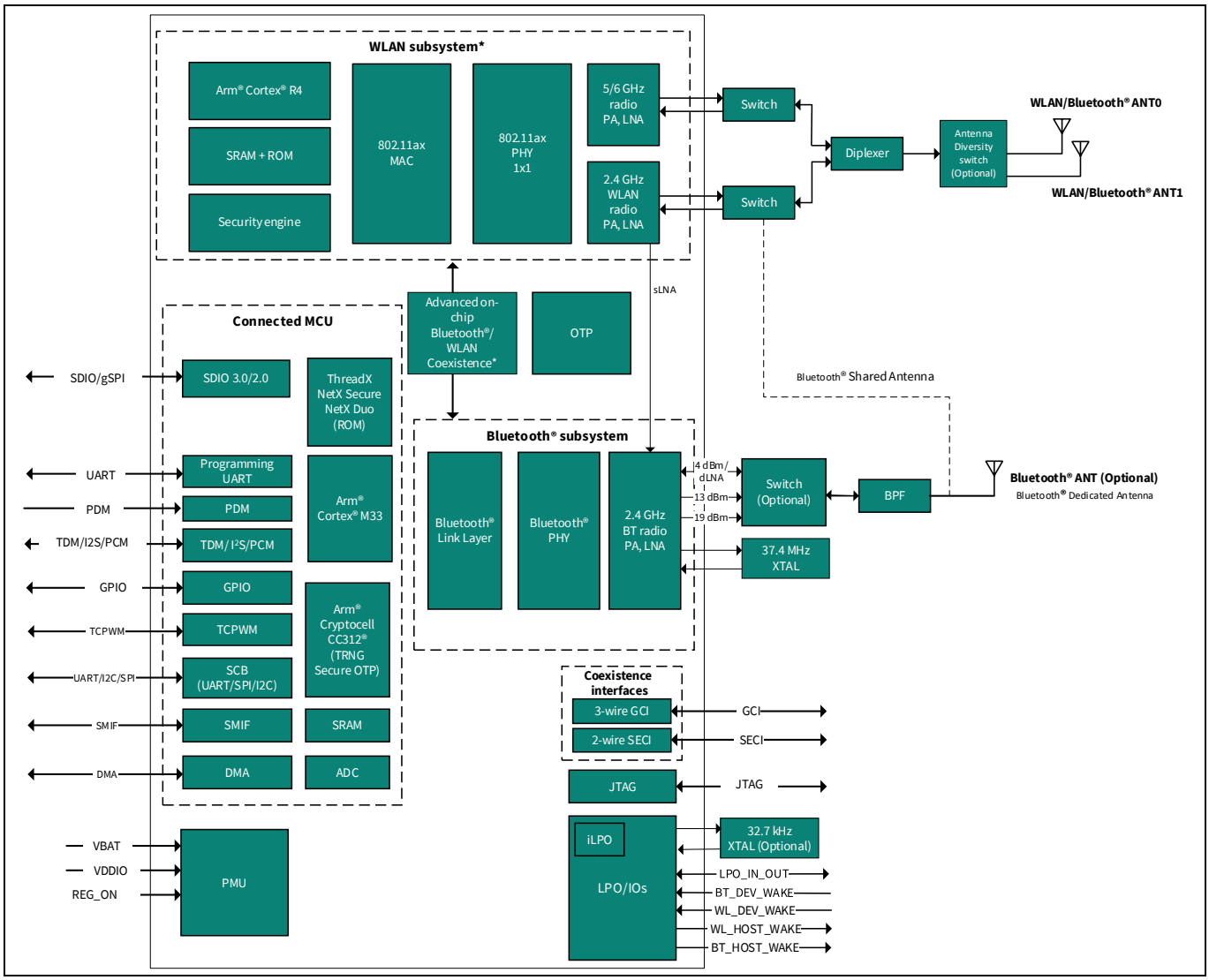


Figure 1 Block diagram

1.4 CYW55913 WLBGA package ball map

Figure 2 shows a CYW55913 WLBGA package ball map.

	R	P	N	M	L	K	J	H	G	F	E	D	C	B	A	
10	WRF_RFIN_5G	WRF_RX5G_GND	WRF_LOGEN_GND	WRF_SYNTH_VDD_1P12	WRF_VCO_GND	NC1	RFSW_CTRL7	VDDIO_RFSW	GPIO_0	GPIO_4	VDDOUT_RF3P3	WLLDO_VDDBAT	BTLDO_VDDBAT	ASR_VDDBAT	ASR_VLX	10
9	WRF_PAOUT_5G	WRF_TX5G_GND	WRF_GPAIO_TSSI2G_OUT				RFSW_CTRL6	VSSC	GPIO_2	GPIO_6	PMU_VDD1P8	VDDOUT_PA3P3	VDDOUT_BT3P3	VDDOUT_CLDO	PVSSA	9
8	WRF_PA_GND	WRF_PA_VDD_3P3	WRF_TX_GND	WRF_SYNTH_VDD_3P3	WRF_SYNTH_GND	NC2	RFSW_CTRL5	VDDC	GPIO_3	GPIO_5	MIC_P	PMU_AVSS	REG_ON	ASR_VDD1P12	REG_ON	8
7	WRF_PAOUT_2G	WRF_TX2G_GND	WRF_LO_GND	WRF_PMU_VDD_3P3	WRF_AFE_VDD_1P12	RFSW_CTRL4	LHL_XTALI	LHL_XTALO	BT_UART_RTS_N	BT_UART_TXD	VSSC	BT_GPIO_5	SDIO_DATA_2	SDIO_DATA_1	SDIO_CLK	7
6	WRF_RFIN_2G	WRF_RX2G_GND	WRF_TSSI5G_OUT	WRF_PMU_GND	WRF_PMU_TX_VDD_1P12	RFSW_CTRL3	VDDC	LHL_GPIO_2	BT_UART_RXD	BT_UART_CTS_N	BT_GPIO_6	VDDIO	SDIO_DATA_3	SDIO_DATA_0	SDIO_CMD	6
5	BTRF_RF_OP	BTRF_VSS	BTRF_IF_VSS	WRF_VDD_1P8	WRF_AFE_LPF_TIA_GND	RFSW_CTRL2	LPO_IN_OUT	LHL_GPIO_3	AVDD_BBPLL	BT_GPIO_4	BT_GPIO_3	VDDC	SMIF_SPHB_DQ3	SMIF_SPHB_DQ2	SMIF_SPHB_CK	5
4	BTRF_VSS	BTRF_TEST	BTRF_PLL_VSS	BTRF_RX_DVSS	RFSW_CTRL1	RFSW_CTRL0	VSSC	LHL_GPIO_4	VSSC	BT_VDDO	BT_GPIO_2	TDM1_DI	SMIF_SPHB_CS0_N	SMIF_SPHB_DQ1	SMIF_SPHB_DQ0	4
3	BTRF_13DBM_OP	BTRF_VCO_VSS		BTRF_LDO_VDD_1P12		JTAG_SEL	BT_DEV_WAKE	LHL_GPIO_5	SMIF_SPHB_CS1_N	TDM1_DO	TDM1_SCK	TDM1_WS	BT_GPIO_17	BT_GPIO_16	BT_GPIO_7	3
2		BTRF_PA_VSS	BTRF_TX_VSS	BTRF_TX_DVSS	XTAL_GND	VDD_XTAL	WL_DEV_WAKE	VDDC	LHL_GPIO_8	BT_HOST_WAKE	TDM1_MCK	TDM2_DI	VSSC	VDDC	TDM2_WS	2
1	BTRF_20DBM_OP	BTRF_PA_VSS	BTRF_PA_VDD_3P3	BTRF_TXIF_VSS	XTAL_XON	XTAL_XOP		LHL_GPIO_9	LHL_GPIO_6	BT_GPIO_0	DMIC_DQ	DMIC_CK	TDM2_DO	TDM2_MCK	TDM2_SCK	1
	R	P	N	M	L	K	J	H	G	F	E	D	C	B	A	

Figure 2 WLBGA package ball map

1.5 Power topology

As shown in Figure 3, Two buck regulators, multiple LDO regulators, and a power management unit are integrated into CYW55913/55912/55911/55903/55902/55901. All regulators are programmable via the PMU. These blocks simplify power supply design for Bluetooth® and WLAN functions in embedded designs. CYW55913/55912/55911/55903/55902/55901 only requires two power supplies, VBAT and VDDIO, to be provided, with all additional voltages being generated by on-chip regulators.

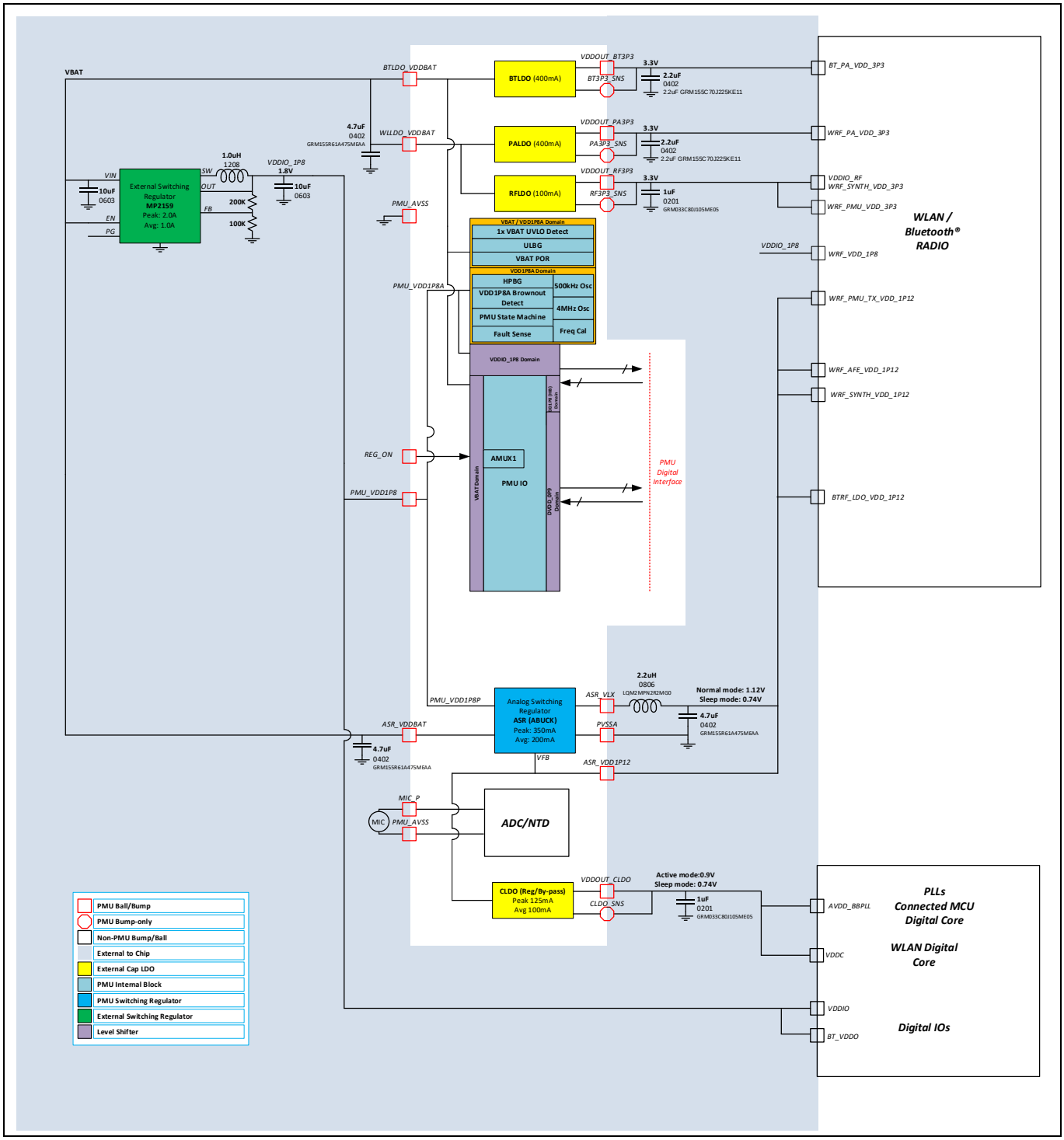


Figure 3 Typical power topology

2 Reference design board

2.1 Block diagram

Note: *Contact Infineon support for the schematic and the Allegro layout file (*.brd). In this document, excerpts from the layout file are provided to show various areas of importance.*

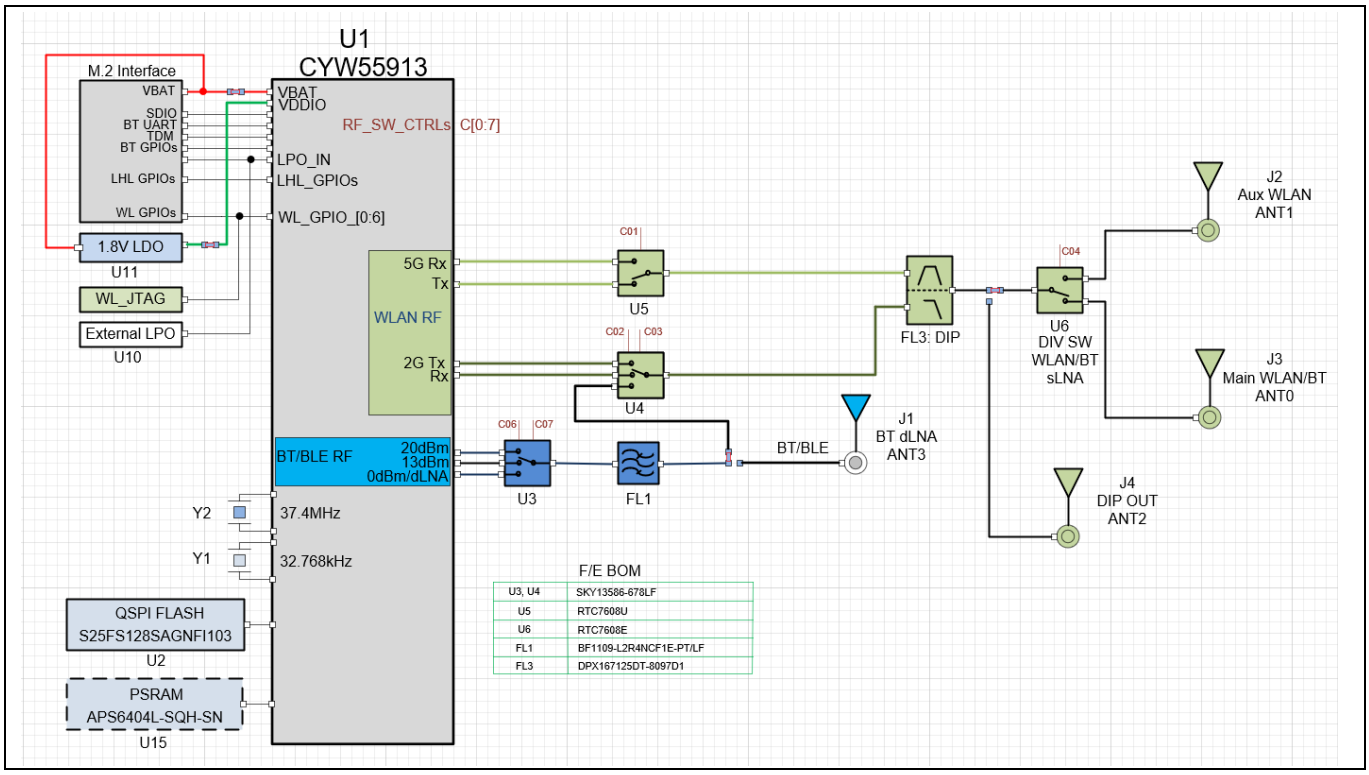


Figure 4 Functional block diagram

2.2 REG_ON Delay circuit requirement

To follow the sequence of VBAT -> VDDIO -> REG_ON for host less scenarios, a delay from VDDIO -> REG_ON is required. This can be achieved via an added RC delay circuit with time constant of approximately 10ms (Figure 5).

- Ensure to tie REG_ON Pins A8 and C8 together.
- Add an RC delay circuit with a pull-up resistor (R7) of 10 kOhm, and pull-down capacitor (R10) of 1 uF.
- If there is a requirement to add more delay or to be more robust and deterministic, a voltage detector circuit needs to be added to meet the same.

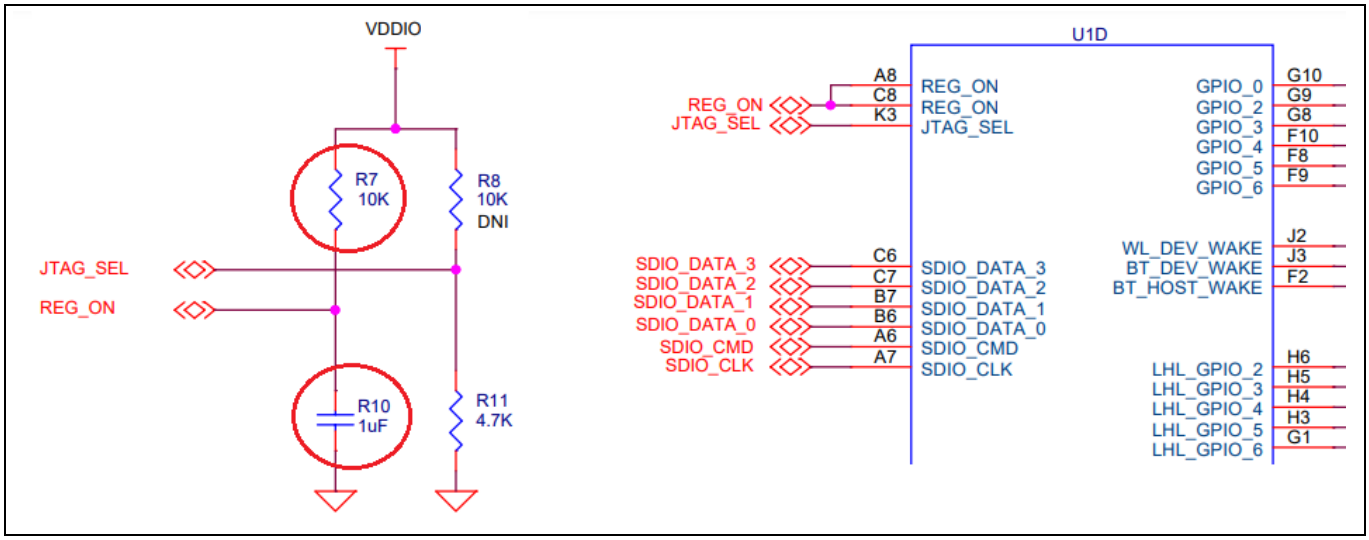


Figure 5 REG_ON delay circuit requirement

2.3 Major components in the reference design

Table 1 Major reference design components

Reference design component designation	Description	Value
C2	ASR VBAT capacitor	4.7 µF
C5	BTLDO VBAT capacitor	4.7 µF
C1	WLLDO VBAT capacitor	4.7 µF
C14	VDDOUT_BT3P3 capacitor	2.2 µF
C11	VDDOUT_RF3P3 capacitor	1 µF
C8	VDDOUT_PA3P3 capacitor	2.2 µF
C3	ABUCK capacitor	4.7 µF
L1	ABUCK inductor	2.2 µH

2.4 Board layers

Note: The screen captures in the following sections are from the reference design layout.

Table 2 Board layers

Layer	Description
1	Component placement, RF signals, and wide power traces
2	Ground plane. Includes ground cut for WLAN traces and antenna connector.
3	Power (Vcc) plane. A few of the power rails are routed at this plane.
4	Ground/Signal plane. A few of the digital signal traces and some power rails are routed at this plane.
5	Ground/Signal plane.
6	Component placement and wide power traces.

2.4.1 Layer 1 (component placement)

Regulators are sensitive to routing parasitics that can cause instability, power efficiency losses, and in extreme cases, loss of voltage regulation. Extreme parasitics due to long and thin traces can create large switching voltage spikes leading to long-term chip reliability problems.

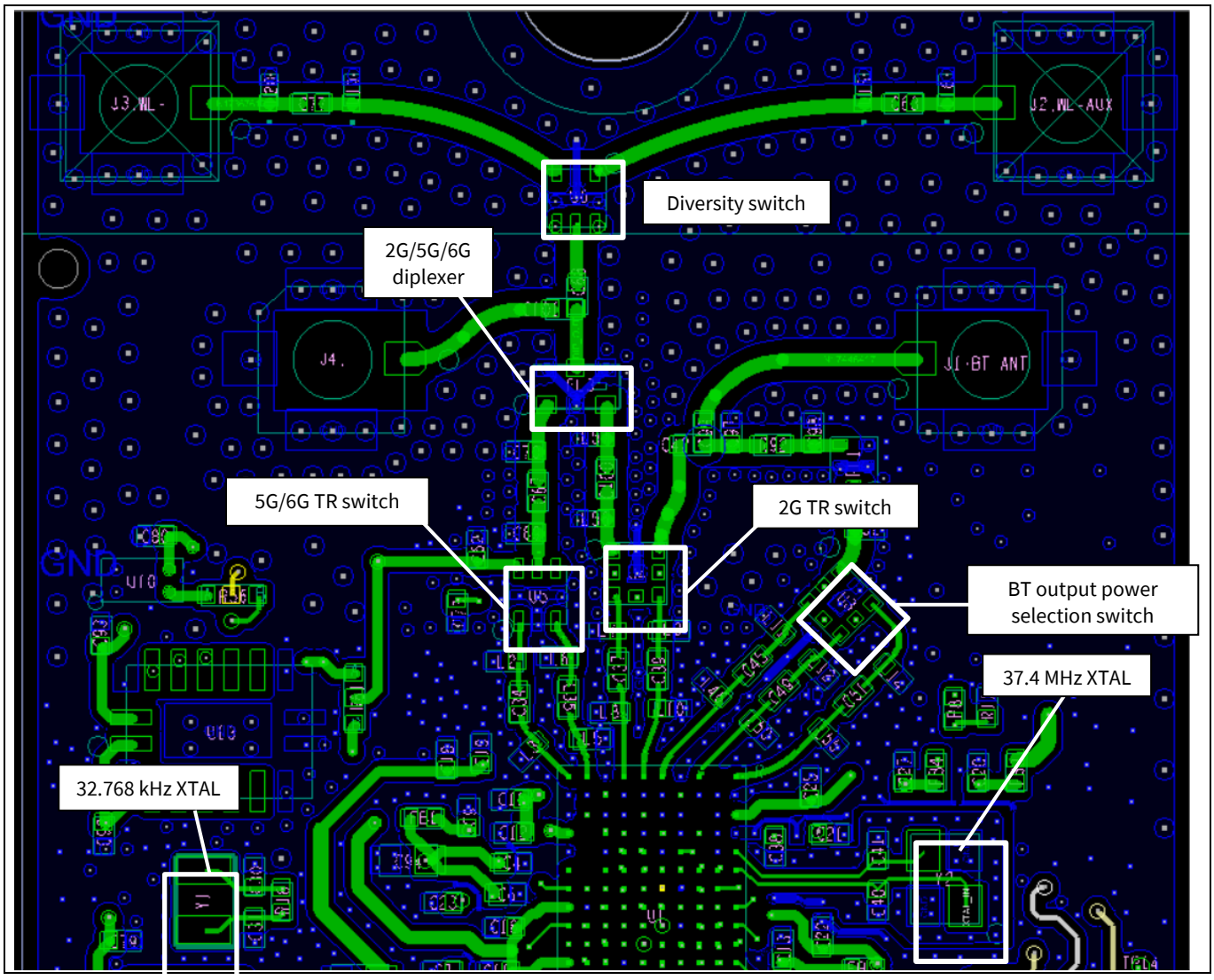


Figure 6 Reference design board Layer 1 (component placement 1 of 2)

Reference design board

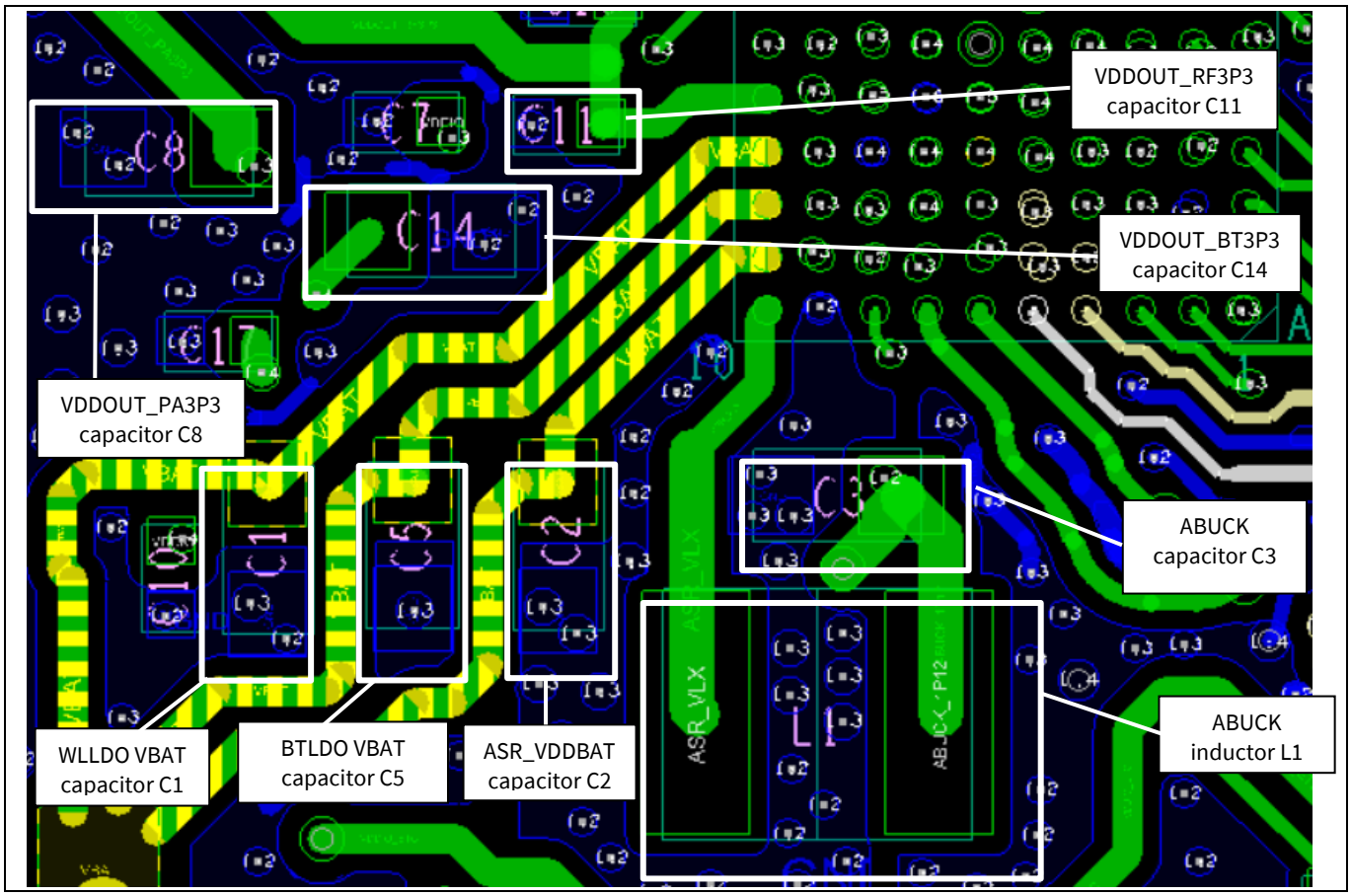


Figure 7 Reference design board Layer 1 (component placement 2 of 2)

2.4.2 Layer 2 (ground plane)

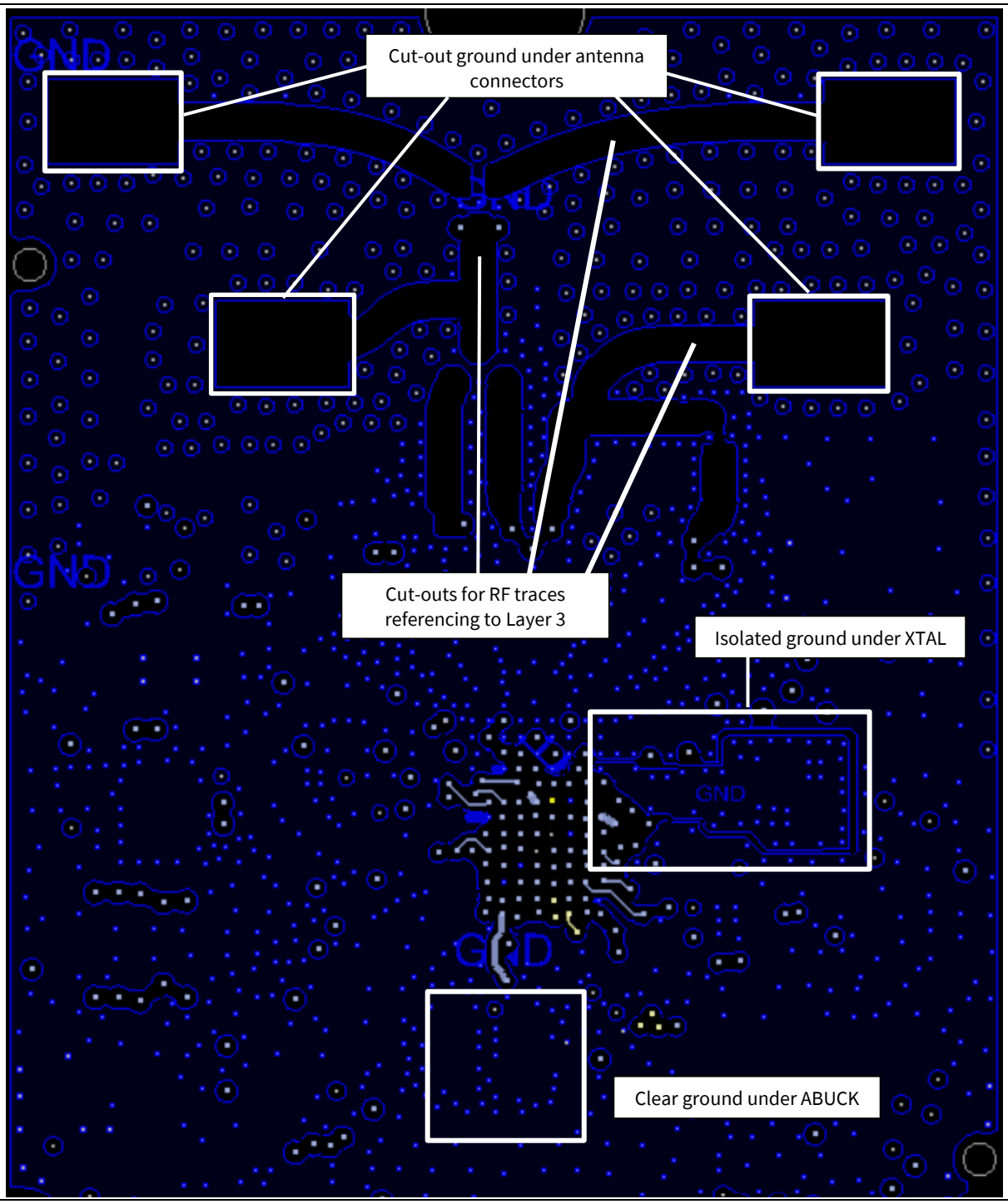


Figure 8 Reference design board layer 2 (ground plane)

2.4.3 Layer 3 (power plane)

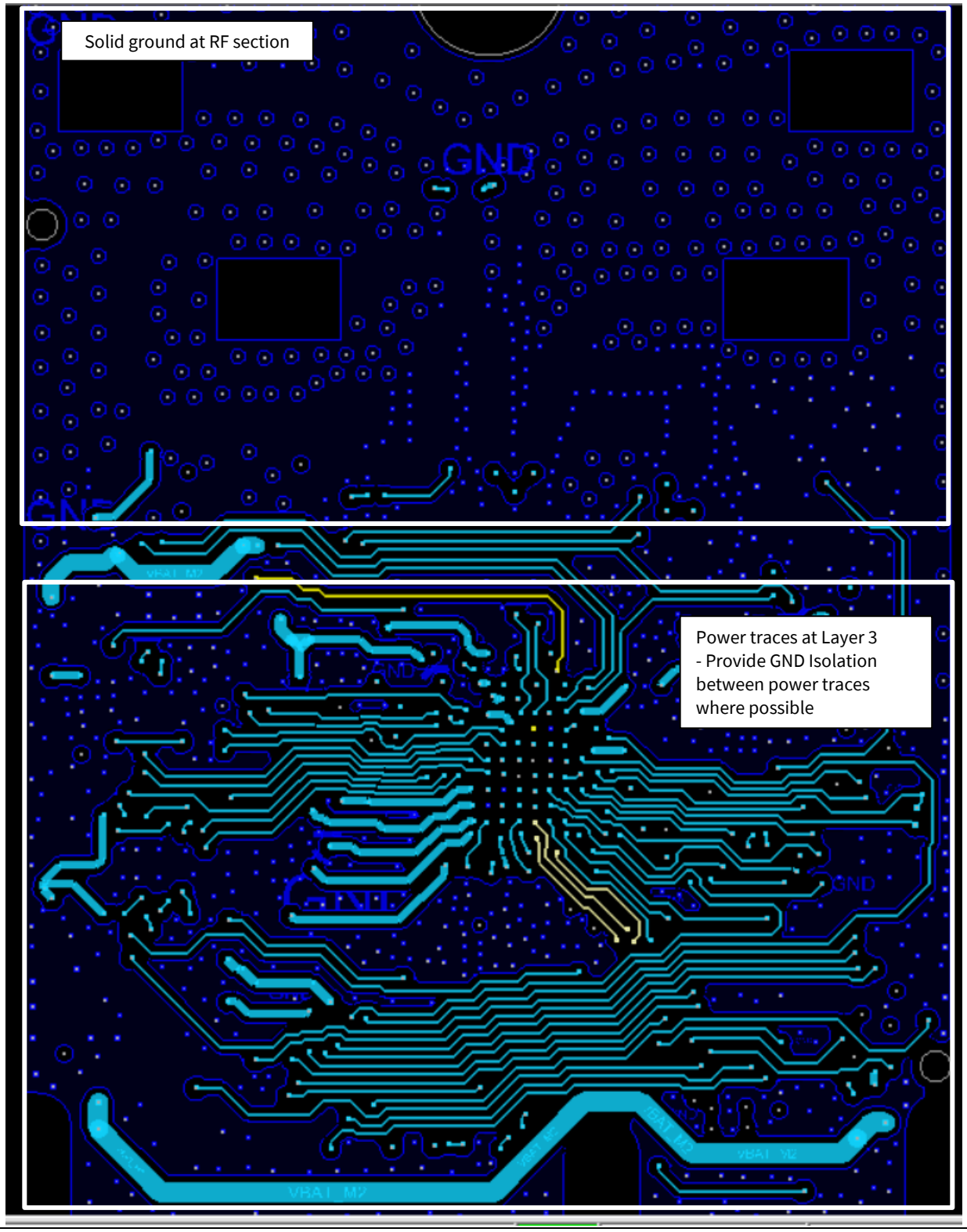


Figure 9 Reference design board Layer 3

2.4.4 Layer 4 (signal/ground plane)

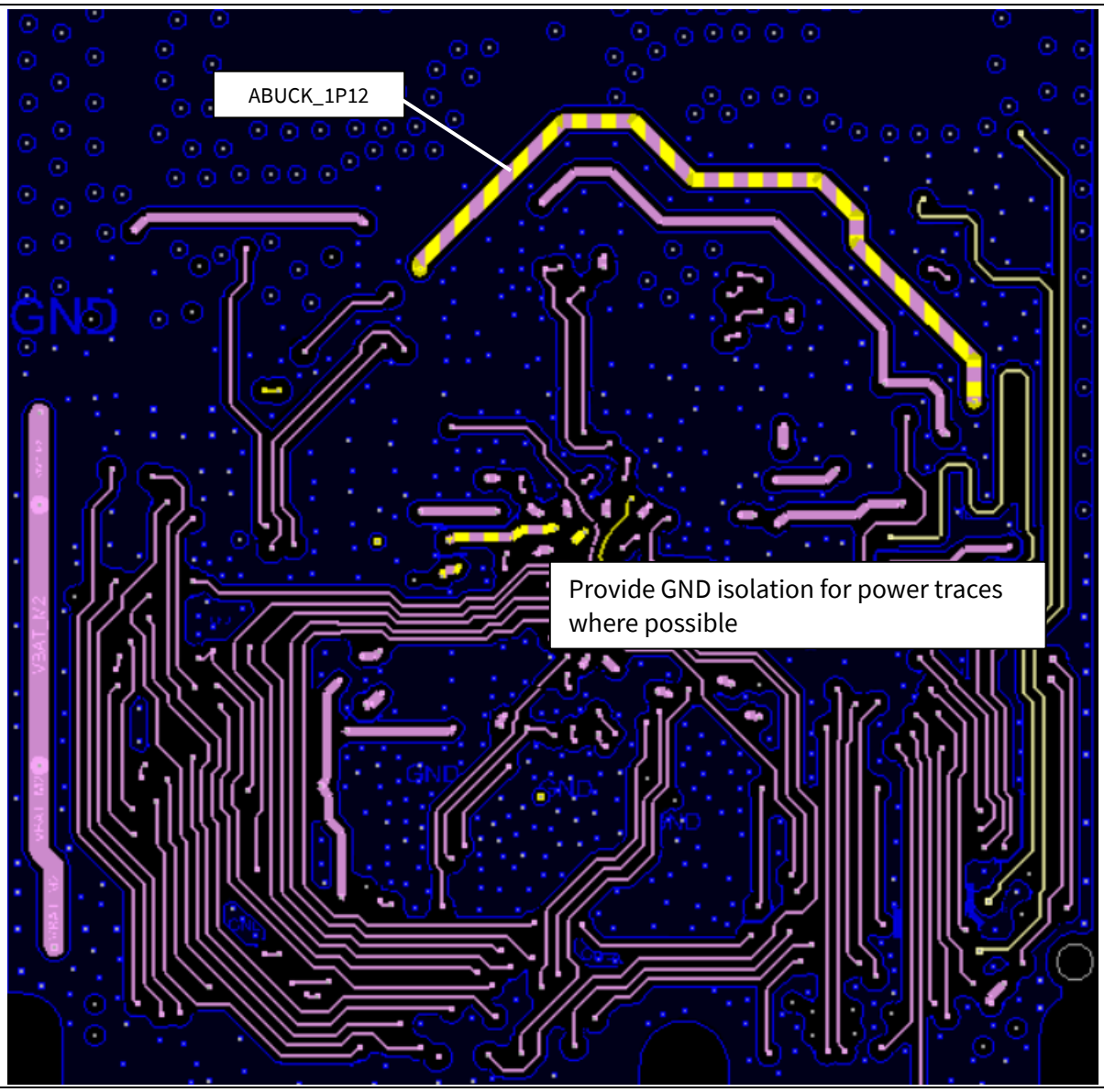


Figure 10 Reference design board Layer 4

2.4.5 Layer 5 (signal plane)

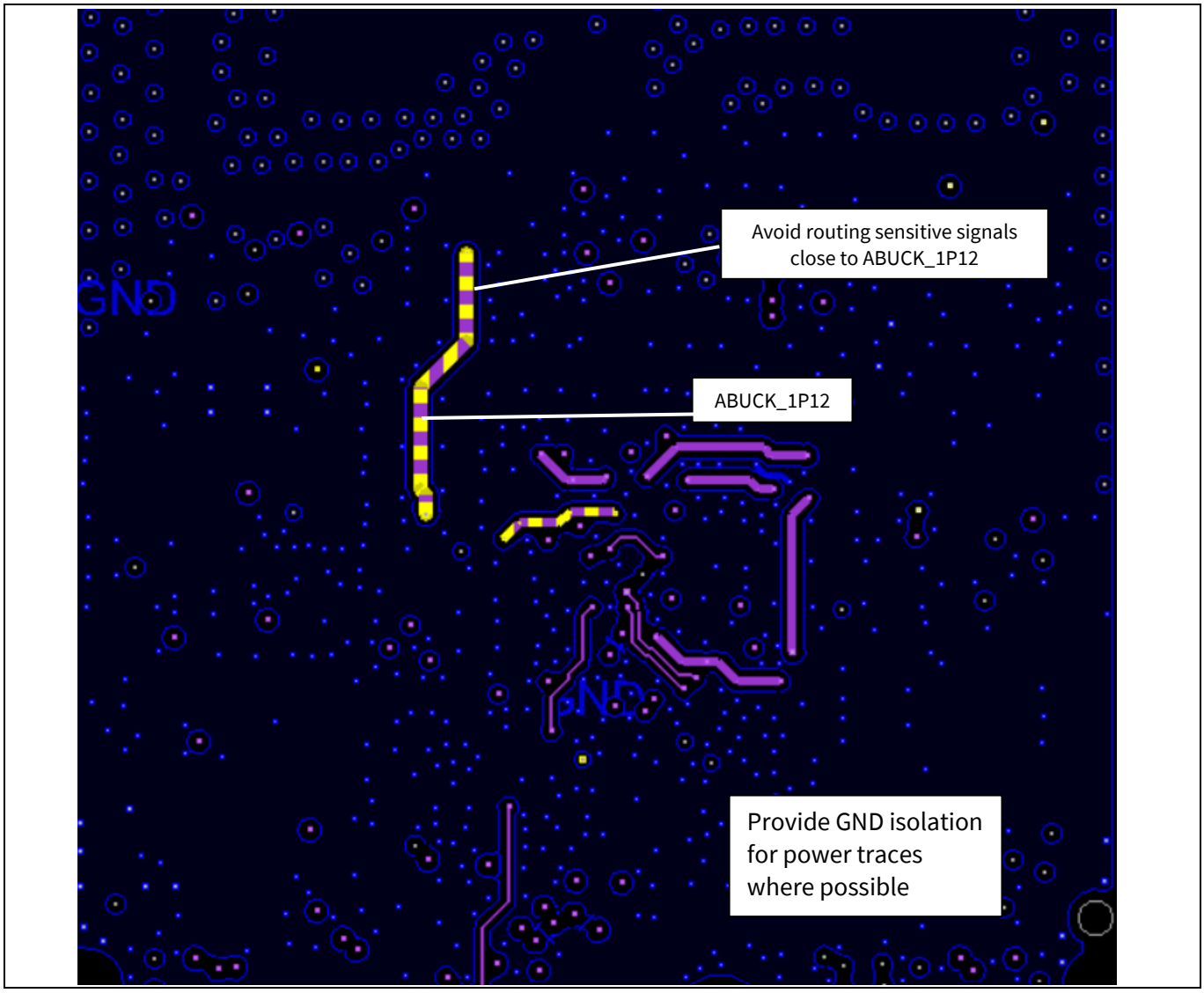


Figure 11 Reference design board layer 5

2.4.6 Layer 6 (signal plane)



Figure 12 Reference design board Layer 6

3 Component selection

This section provides component selection guidance for optimizing the performance of the following power rails:

- VBAT (ASR_VDDBAT, BTLDO_VDDBAT, WLLDO_VDDBAT)
- ABUCK (ASR_VLX)
- VDDOUT_PA3P3
- VDDOUT_RF3P3
- VDDOUT_BT3P3
- VDDOUT_MISC
- Front end

3.1 External components selection

The choice of inductor depends on the DC resistance (DCR), AC resistance (ACR), saturation currents, temperature rise criteria, and footprint/height. The use of smaller components may negatively impact overall efficiency performance. Ensure that the effective capacitance does not drop by more than 69 percent of the nominal value at 3.6 V.

3.2 ABUCK inductance

The nominal inductor value is 2.2 µH.

Table 3 Recommended inductors

Vendor	Part #	L (µH)	Tol (%)	DCRmax (mΩ)	DCRAVG (mΩ)	Current rating (A)	Size (in)	Purpose
Murata	DFE201610E_2R2	2.2	±20	140	117	1.7	0806	ASR output inductor
Murata	LQM2MPN2R2MG0	2.2	±20	138	110	1.2	0806	
Murata	WIP201610S-	2.2	±20	140	117	2.0	0806	

Table 4 ASR (ABUCK) generic inductor specifications

Requirement	Value
Typical value	2.2 µH
PWM switching frequency	4 MHz
Tolerance	±20%
DCR nominal	0.11 Ω to 0.15 Ω
ACR at 4 MHz	≤ 1.2 Ω at no-load
Maximum rated current (40°C rise)	1600 mA
Inductance vs. DC bias-current profile	2.0 µH at 200 mA 1.9 µH at 400 mA 1.2 µH at 600 mA 0.9 µH at 800 mA 0.7 µH at 1000 mA

The inductor characteristics listed in the following table must be known in order to determine the suitability with the ASR (ABUCK).

Table 5 ASR (ABUCK) inductor requirements

Criteria	Value
Inductor tolerance	No greater than ±20%
DCR	DCR versus temperature
ACR	• ACR vs. frequency under no-load, mid-load, and maximum load conditions • ACR vs. frequency under minimum, typical, and maximum operating temperatures, at maximum load • ACR vs. 4 MHz ripple current amplitude under no load and maximum load conditions
I_{SAT}	Saturation current based on (nominal inductance of 30%)
I_{TEMP}	• Temperature saturation current is specified when the self-heating temperature of the inductor is raised +40°C • Inductance under I_{TEMP} versus operating temperature range

3.2.1 Inductor comparative criteria

3.2.1.1 DC resistance

The DC resistance (DCR) contributes to the conduction power losses of the ABUCK. Excessive DCR can cause power losses up to 5%.

3.2.1.2 Saturation current (I_{SAT})

The saturation current (ISAT) is defined as the load current that causes the inductance to drop by 30% of the nominal value. Typically, datasheets of multilayered inductors do not explicitly present the saturation current information because of the lower current level.

An excessive drop in inductance can lead to additional inductor losses because of increased inductor ripple current through the ACR. Increased inductor ripple current can also lead to greater switching-current injection into ground.

For pulse width modulation (PWM) loop stability, effective inductance at a 600-mA load should be at least 1.2 µH across all parts (tolerance included). For optimal peak PWM efficiency at a 200-mA load, the effective inductance should be at least 1.9 µH.

3.2.1.3 Temperature saturation current (I_{TEMP})

The temperature saturation current (I_{TEMP}) is the current that causes the inductor to self-heat above +40°C. Typically, I_{TEMP} is provided in the datasheets and is usually higher than the I_{SAT} saturation current. I_{TEMP} should be at least 1000 mA.

3.2.1.4 Inductance under I_{TEMP} vs. operating temperature range

Take the following information into account when considering an inductance under I_{TEMP} over the entire operating temperature range:

- Characterize the inductance assuming an I_{TEMP} temperature saturation current across the full operating temperature range of the inductor.
- The recommended operating temperature range is from –40°C to +125°C, including the rise in temperature due to inductor self-heating (+40°C).
- The inductor operating temperature (up to +125°C including inductor self-heating of +40°C) corresponds to the customer product temperature maximum of 85°C.
- The inductance should not vary by more than ±4%.

3.3 External bypass capacitance

Table 6 Capacitor recommendations

Vendor	Part	C (µF)	Tol (%)	Voltage (V)	Temp rating	Size (in)	Purpose
Murata	GRM033R60J224ME15	0.22	±20	6.3	X5R	0201	PMU input bypass
Murata	GRM033R60J105MEA2	1.0	±20	6.3	X5R	0201	CLDO, RFLDO output
Murata	GRM033C80J105ME05	1.0	±20	6.3	X6S	0201	
Murata	GRM155C70J225KE11	2.2	±10	6.3	X7S	0402	BTLDO, PALDO outputs
Murata	GRM155R61C225KE11	2.2	±10	16	X5R	0402	
Murata	GRM155R61A475MEAA	4.7	±20	10	X5R	0402	ASR, PMU input bypass
Murata	GRM188R60J475ME84	4.7	±20	6.3	X5R	0603	
Murata	GRM188R60J475KE19	4.7	±10	6.3	X5R	0603	
Murata	GRM188R61C106KAAL	10	±10	16	X5R	0603	

Three critical parts used on the WLAN and Bluetooth® front end are diplexer, RF switches, and band-pass filters.

4 Routing guidelines

4.1 General PCB design guidelines

- Keep noisy lines and crystal trace away from RF traces and place ground vias along RF traces.
- Do not use ground cuts under RF traces.
- Properly ground RF switches using vias.
- Do not place noisy supply lines (such as VBAT and VDDIO) or noisy digital lines (UART, Flash and SDIO) over sensitive supply traces like radio power rails.
- Minimize the coupling of noisy signals to crystal traces.
- Avoid different power traces on adjacent layer overlap.
- Separate different power traces on the same layer as far as possible.
- Add ground side shielding to improve isolation to different neighboring power traces.
- Avoid placing crystal-trace vias close to noisy signals on other layers.
- Use a solid ground layer under the ABUCK regulator.
- Use power supply traces wide enough to support required currents. For example, 10 mils, ½ oz. copper trace can handle 500 mA (at 10°C). The width must be increased to 25 mils for 900 mA. The current-carrying capacity goes up to 700 mA for 10 mils and 1.5 A for 25 mils at 30°C for the same ½ oz. copper weight.
- When trace routing jumps layers, use multiple vias if the trace needs to support a high current.

Note: *In several figures throughout this section, a star symbol is used to indicate a star connection. A star connection is formed at a decoupling capacitor that attaches to more than one circuit element.*

4.2 Routing the VBAT and ABUCK networks

The main power supply (VBAT) coming directly from the platform battery, and the switching regulator (ABUCK) network generating 1.12 V are shown below.

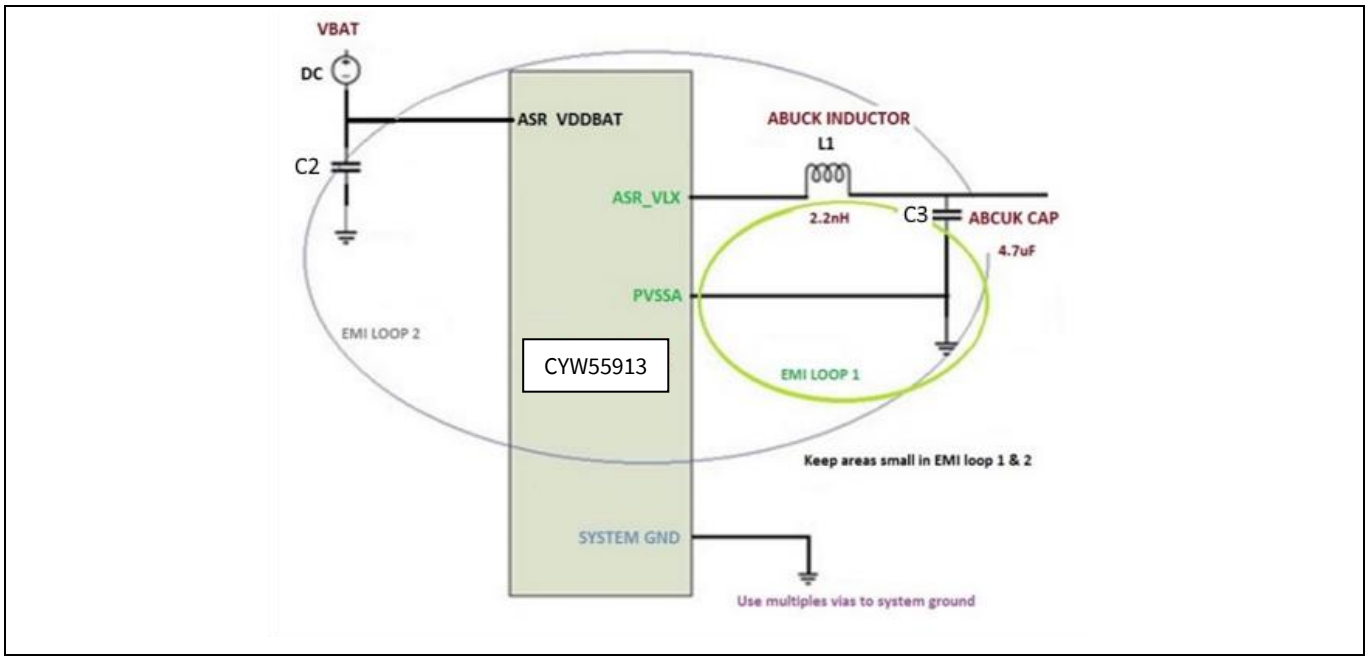


Figure 13 VBAT and ABUCK power networks

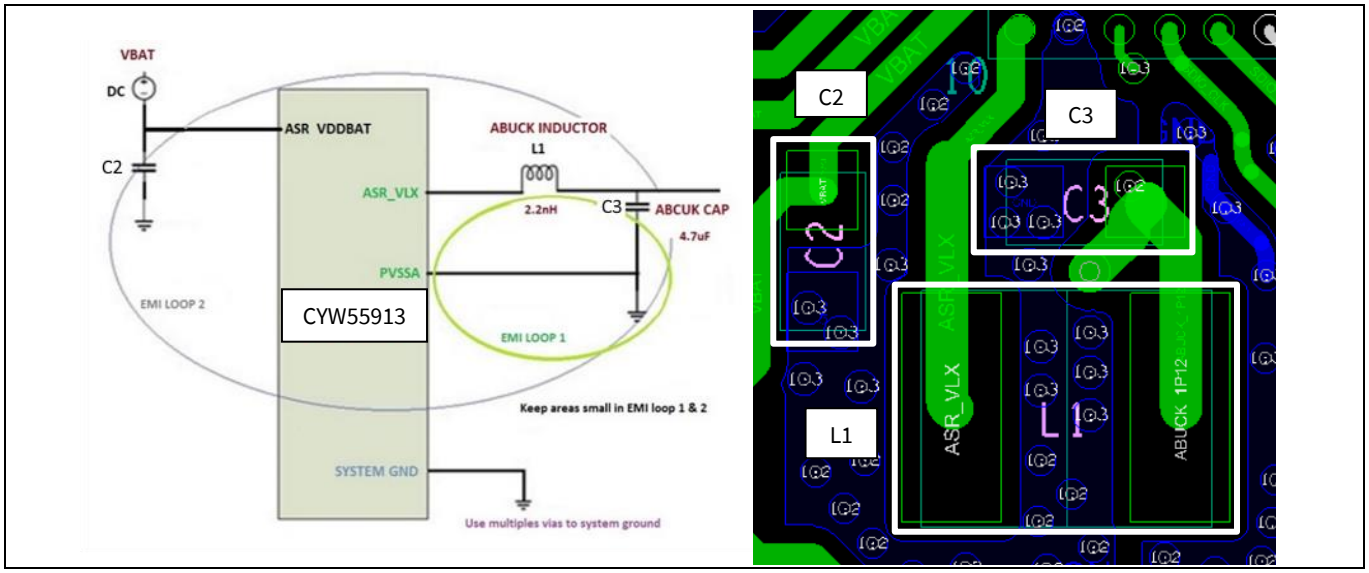


Figure 14 VBAT and ABUCK power supply networks – a closer look

The following routing and placement guidelines apply to the VBAT capacitor, the ABUCK capacitor, the ABUCK inductor, and their associated traces:

- Place all components on the top layer close to pins A9, A10, and B10.
- To handle the large currents between the previously mentioned components and pins, use traces that are as thick as possible.
- Use multiple vias to the second layer to reduce the area of EMI loops 1 and 2.
- To prevent ABUCK circuit transients from coupling onto lower-layer traces, use a solid (layer 2) ground under the ABUCK inductor. See the following diagram for the ground used under the ABUCK inductor in the reference design.

Note: The peak current flowing through the ABUCK inductor and capacitor is 400 mA. To avoid large switching-voltage spikes that lead to power efficiency losses, long-term chip reliability problems, or both, do not use thin and long traces to these components.

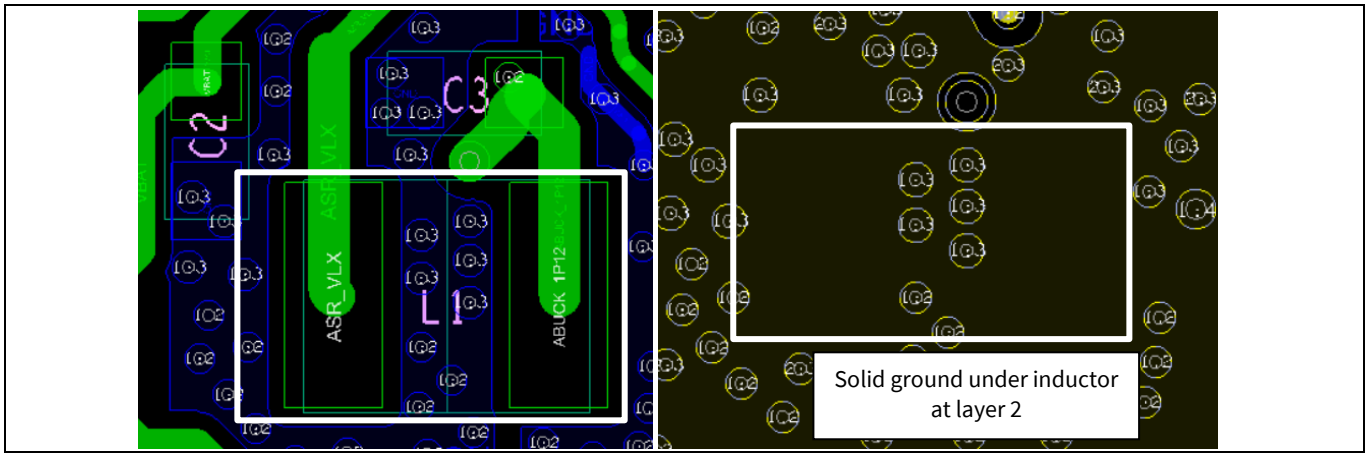


Figure 15 ABUCK inductor (and capacitor) ground plane

Routing guidelines

The following diagram shows the EMI loops in the ABUCK switching regulator. Sensitive signals should not be routed through the areas bounded by the two loops that go through the power MOSFET switches of the ABUCK. Each loop starts at a capacitor and ends at its associated ground terminal. The area within each loop is proportional to the EMI.

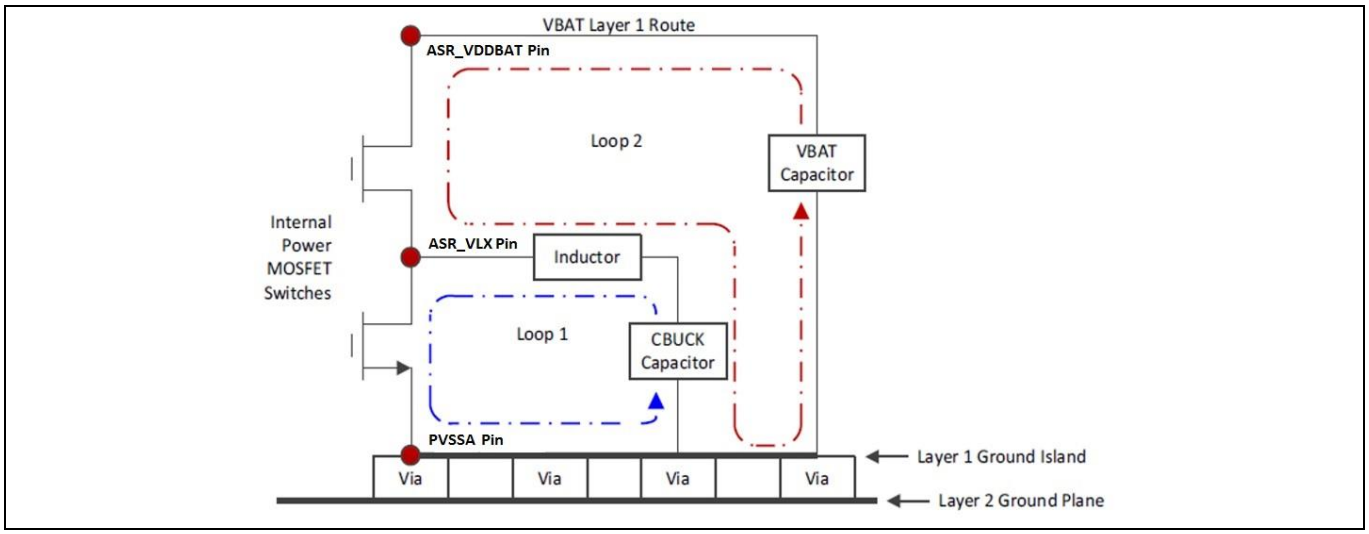


Figure 16 EMI loops in buck switching regulator

When addressing EMI, consider the following:

- Avoid routing sensitive signals through the areas bounded by these two loops.
- The VBAT and ABUCK capacitors sit on a top-layer ground island directly connected to the PVSSA pin.
- Multiple vias are used to connect the ground island on layer 1 to the ground plane on layer 2.
- The ground plane on layer 2 also acts as a shield over the two EMI loops to prevent noise radiation from reaching the adjacent layers.

Additional important points:

- Bypass capacitors on VBAT and ABUCK should have as many ground vias as possible.
- Isolate the ground plane around ABUCK to avoid noise coupling.
- Use at least two vias for the bypass capacitor.
- Minimize the area enclosed by the following loops to reduce EMI in the forward and return paths:
  - ASR_VLX to inductor and capacitor to PVSSA
  - ASR_VDDBAT to supply bypass capacitors to PVSSA

4.3 Routing the VDDOUT_RF3P3 network

Note:

1. *Power traces are routed as a star configuration.*
2. *If your design allows the component to be placed at the bottom, place the decoupling capacitor at the bottom close to the power input pin as necessary, so that the decoupling capacitor is more effective.*

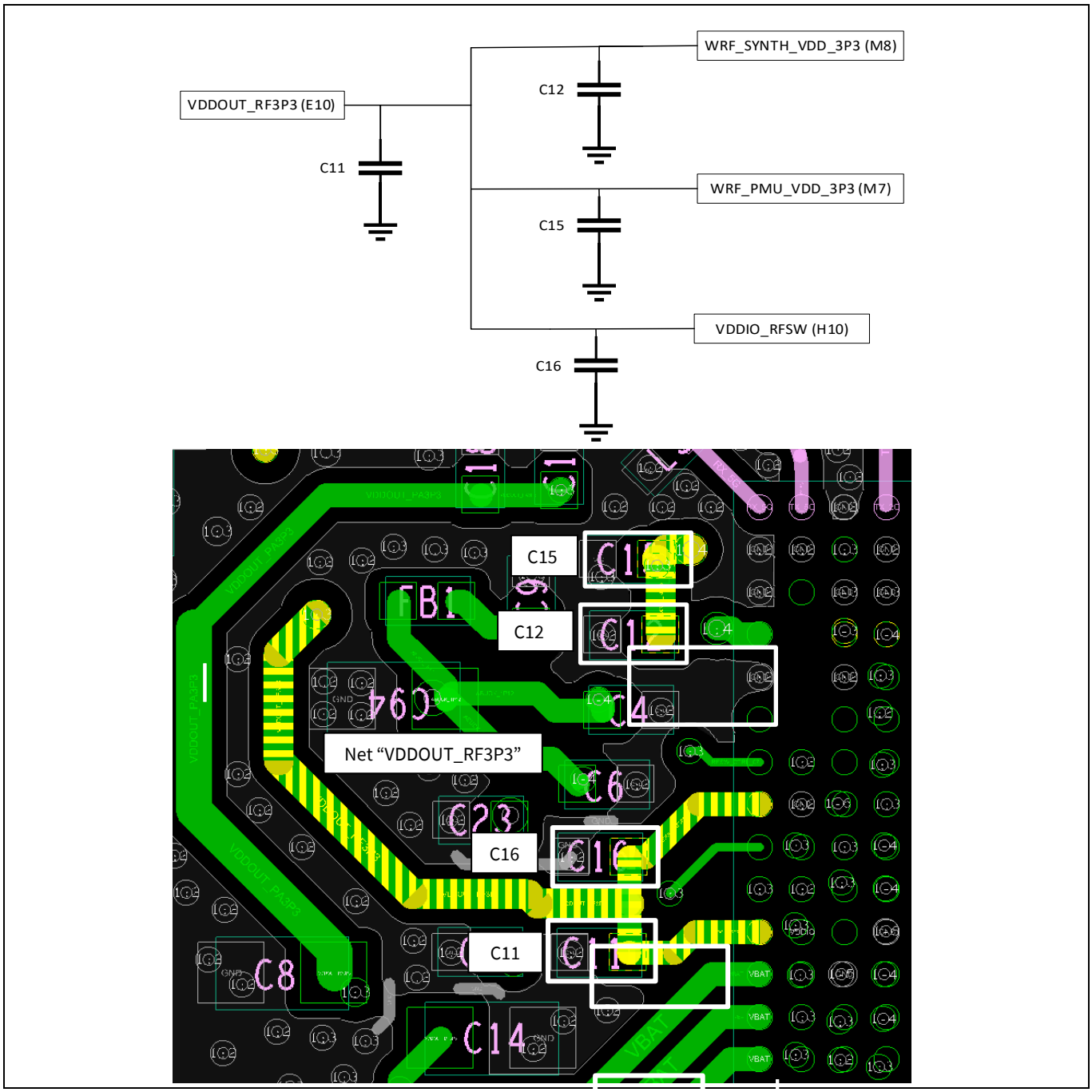


Figure 17 VDDOUT_RF3P3 network (top layer)

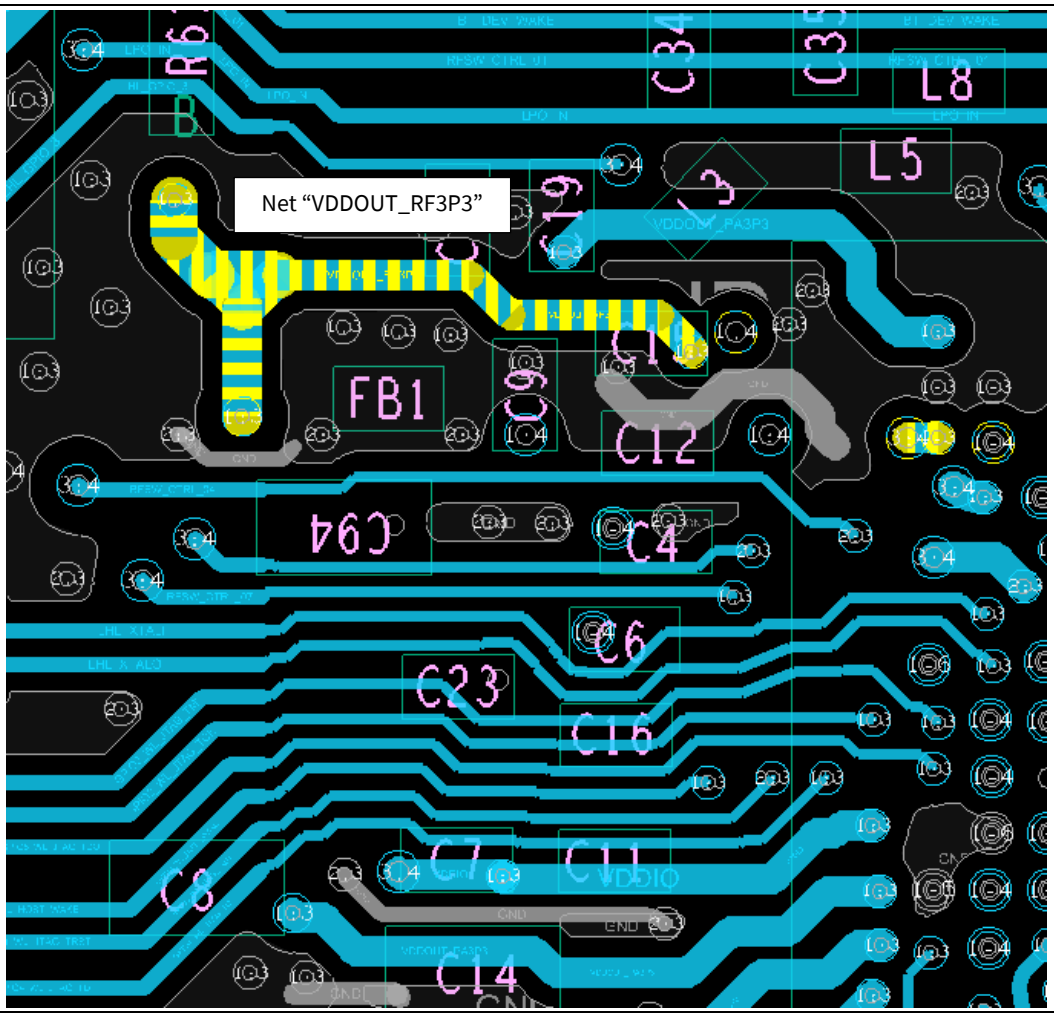


Figure 18 FiVDDOUT_RF3P3 network (layer 3)

4.4 Routing the VDDOUT_PA3P3 network

Note:

1. *If your design allows the component to be placed at the bottom, place the decoupling capacitor at the bottom close to the power input pin, so that the decoupling capacitor is more effective.*
2. *If a power rail requires multiple decoupling capacitors, place the smaller value capacitor closer to the CYW55913 chip.*
3. *One power amplifier (PA) in each core can draw up to 500 mA current; therefore, the trace width should be wide enough to support high current.*
4. *Avoid routing other power signals near VDDOUT_PA3P3.*

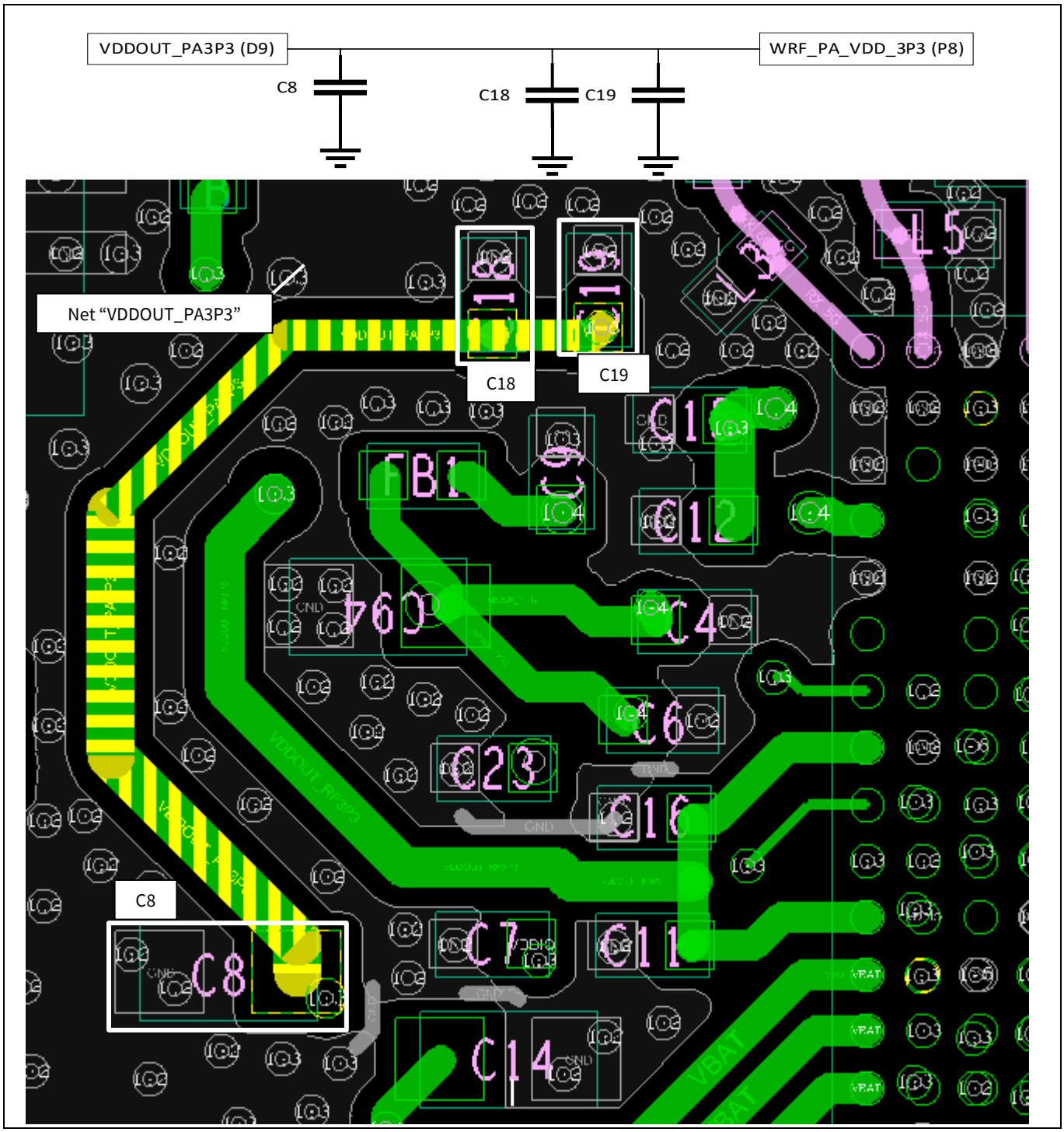


Figure 19 VDDOUT_PA3P3 network (top layer)

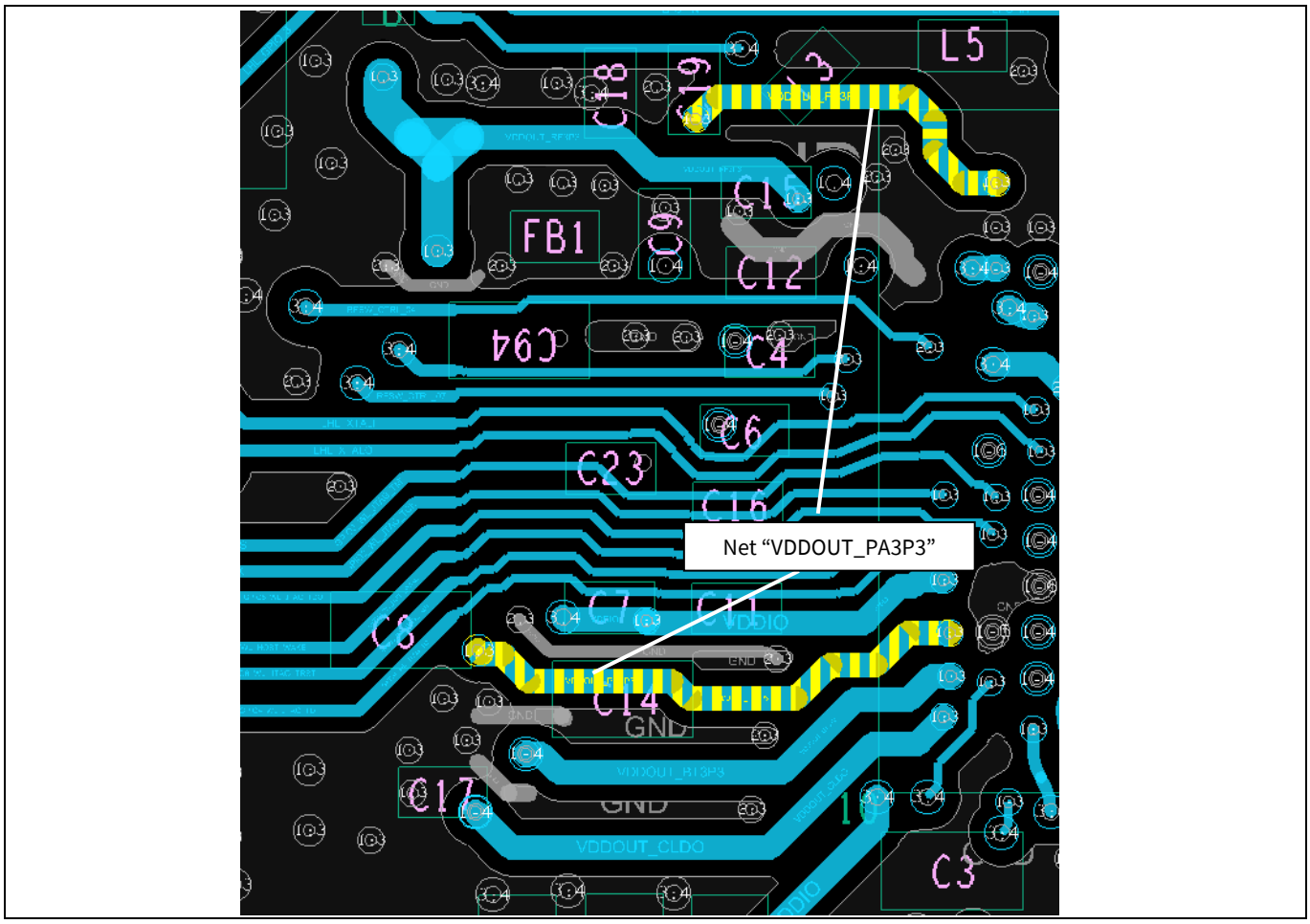


Figure 20 VDDOUT_PA3P3 network (layer 3 and layer 5)

4.5 Routing the ABUCK_1P12 network

Note:

1. *Power traces are routed as a star configuration.*
2. *A few power rails have two decoupling capacitors: if your design allows the component to be placed at the bottom, place the decoupling capacitor at the bottom close to the power input pin, so that the decoupling capacitor is more effective. Place the smaller value capacitor closer to the CYW55913 chip.*

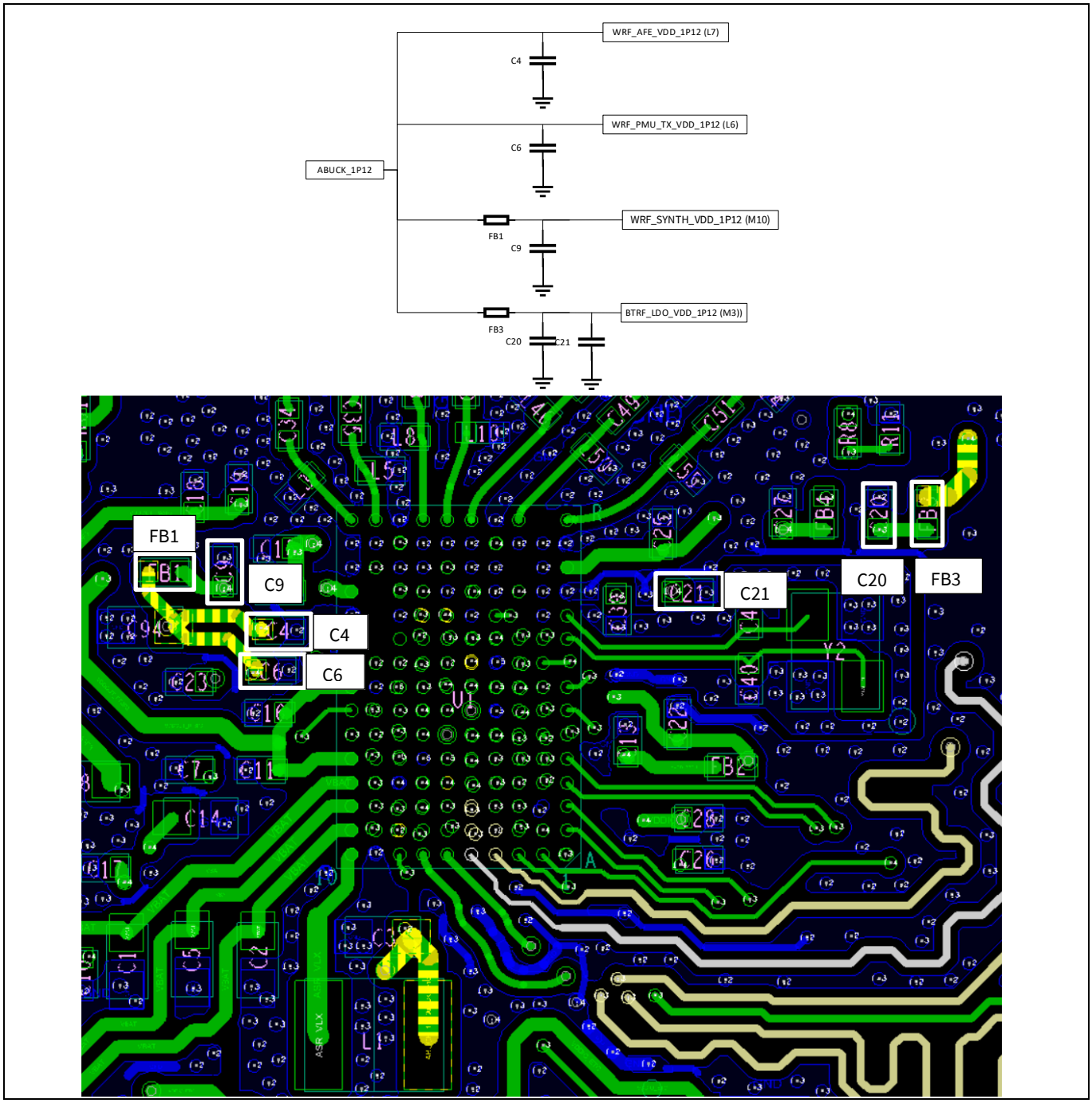


Figure 21 ABUCK_1P12 network (top layer)

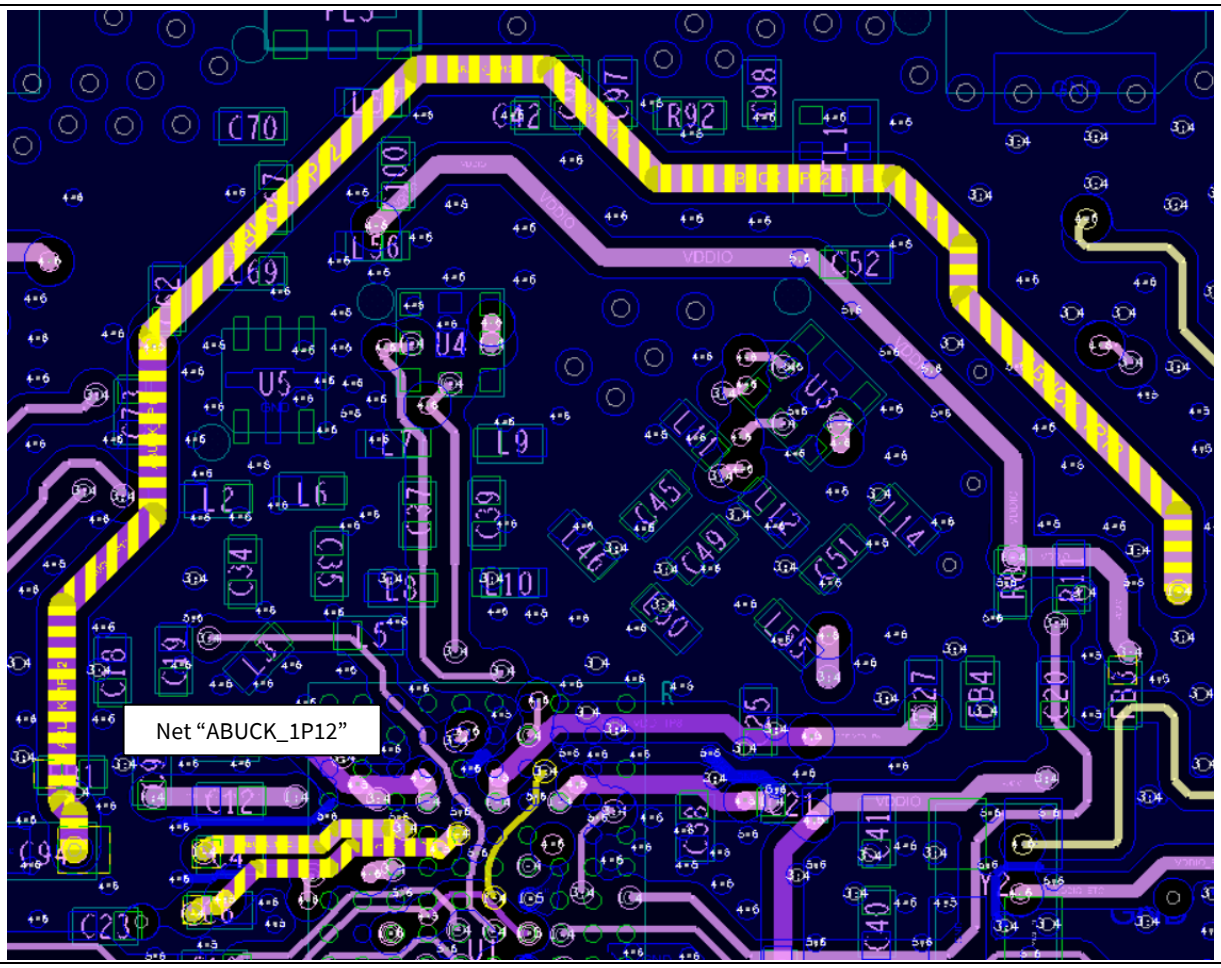


Figure 22 ABUCK_1P12 network (layers 4 and 5)

4.6 Routing the VDDOUT_BT3P3 trace

Note: *BTRF_PA_VDD_3P3 has two decoupling capacitors, but C24 is DNI by default. If your PCB space is limited, C24 can be omitted.*

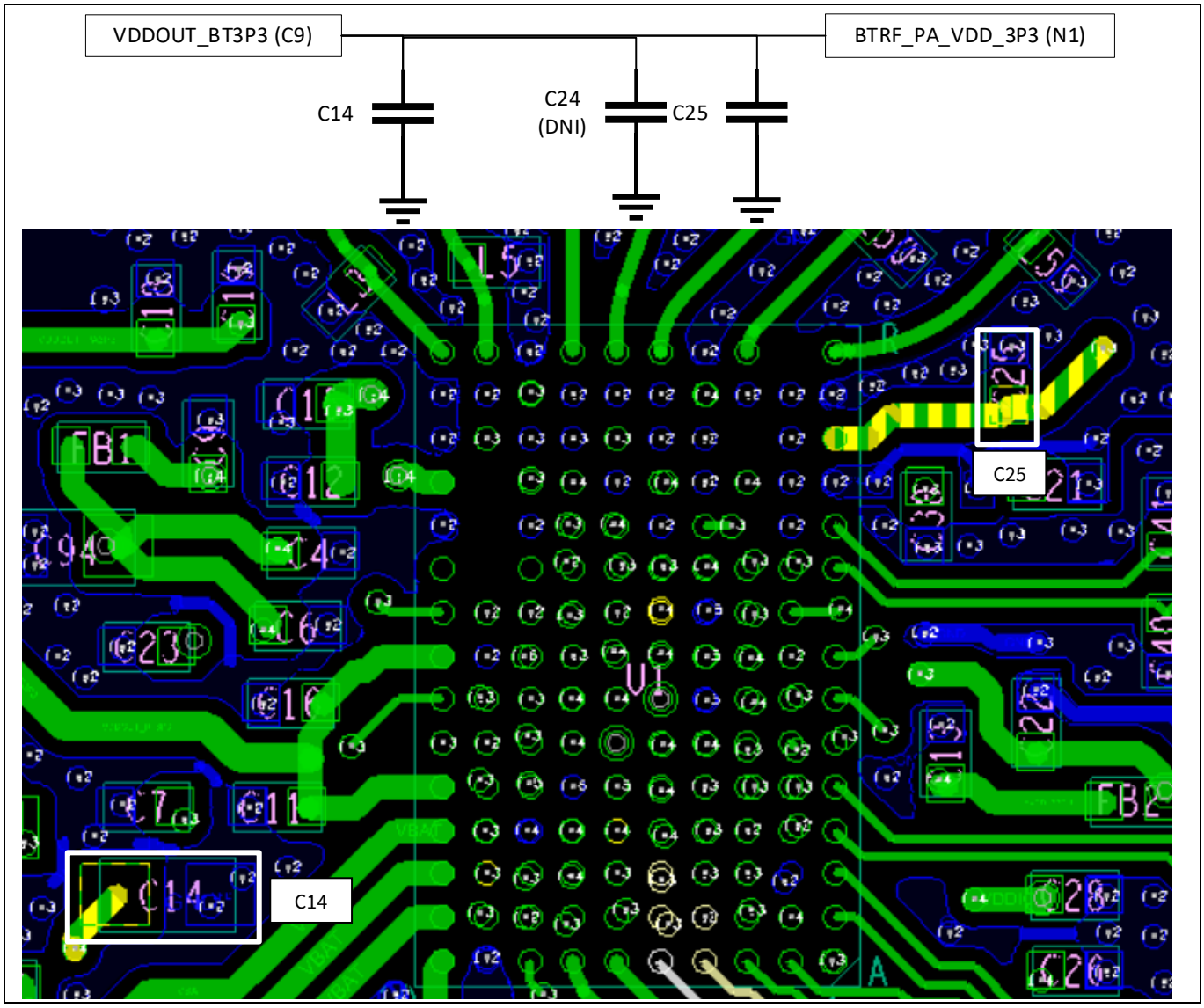


Figure 23 VDDOUT_BT3P3 routing (top layer)

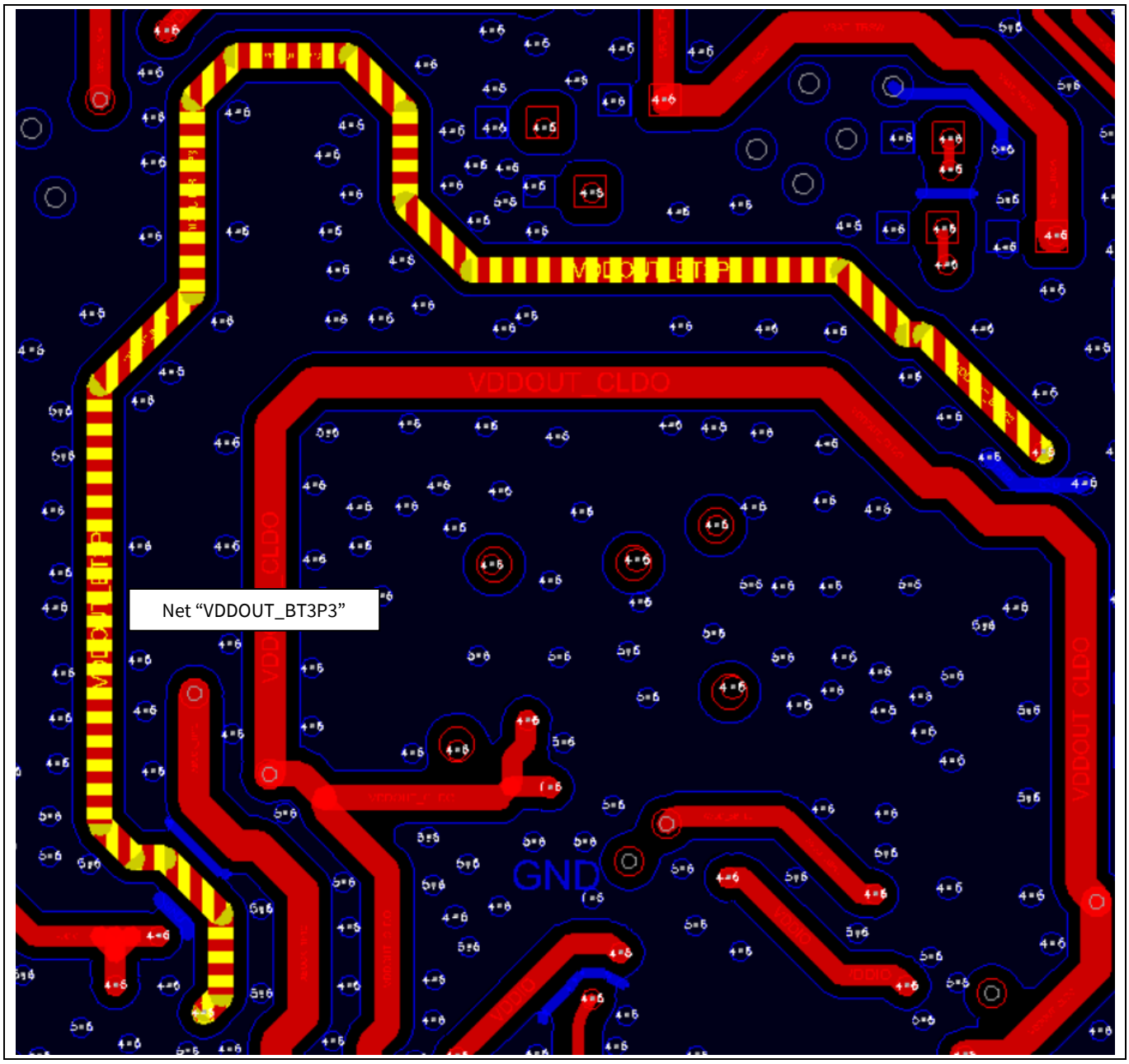


Figure 24 VDDOUT_BT3P3 routing (bottom layer)

4.7 Routing the VDDOUT_CLDO traces

Note:

1. *Power traces are routed as a star configuration.*
2. *A few power rails have two decoupling capacitors: if your design allows the component to be placed at the bottom, place the decoupling capacitor at the bottom close to the power input pin, so that the decoupling capacitor is more effective. Place the smaller value capacitor closer to the CYW55913 chip.*

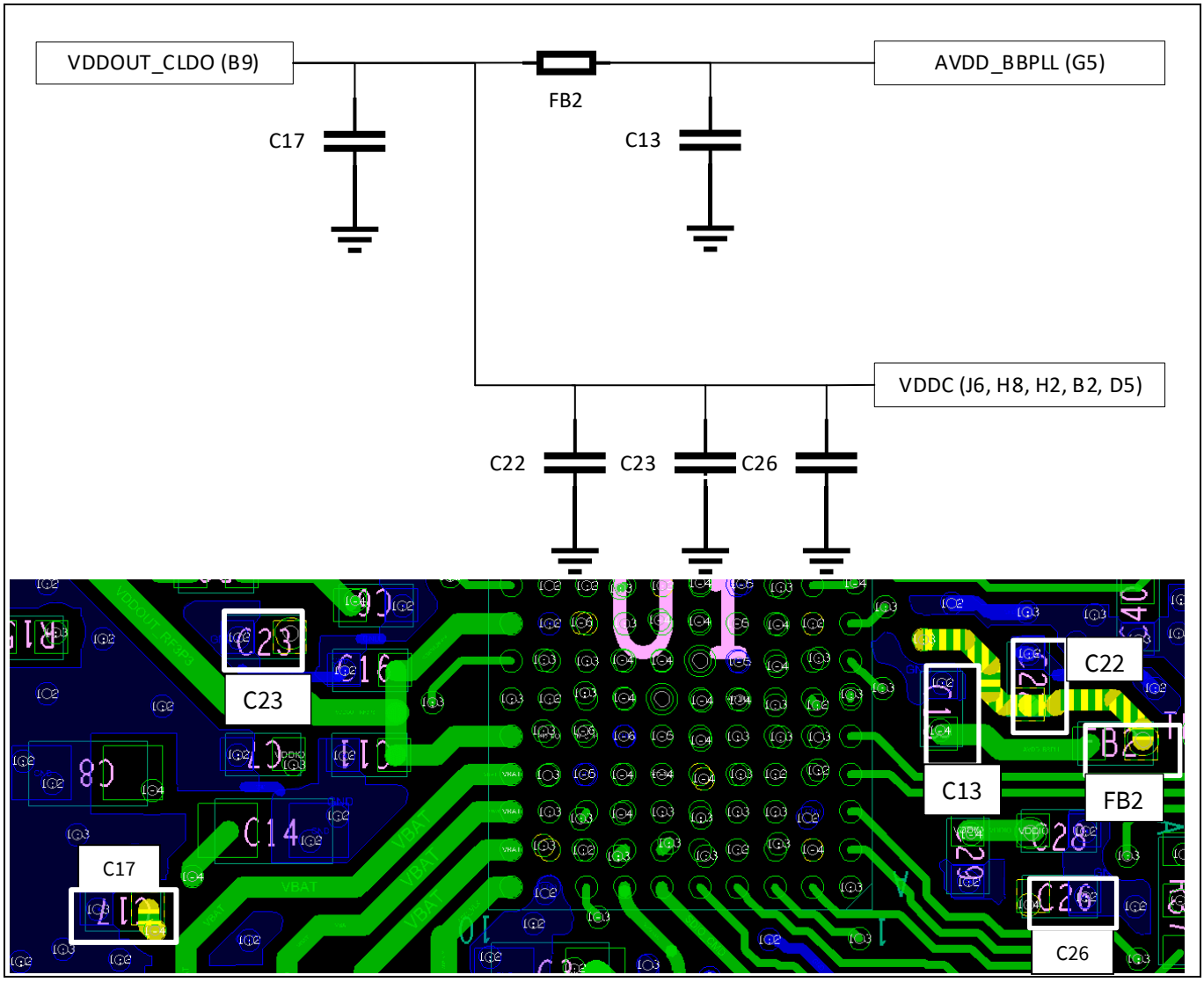


Figure 25 VDDOUT_CLDO routing (top layer)

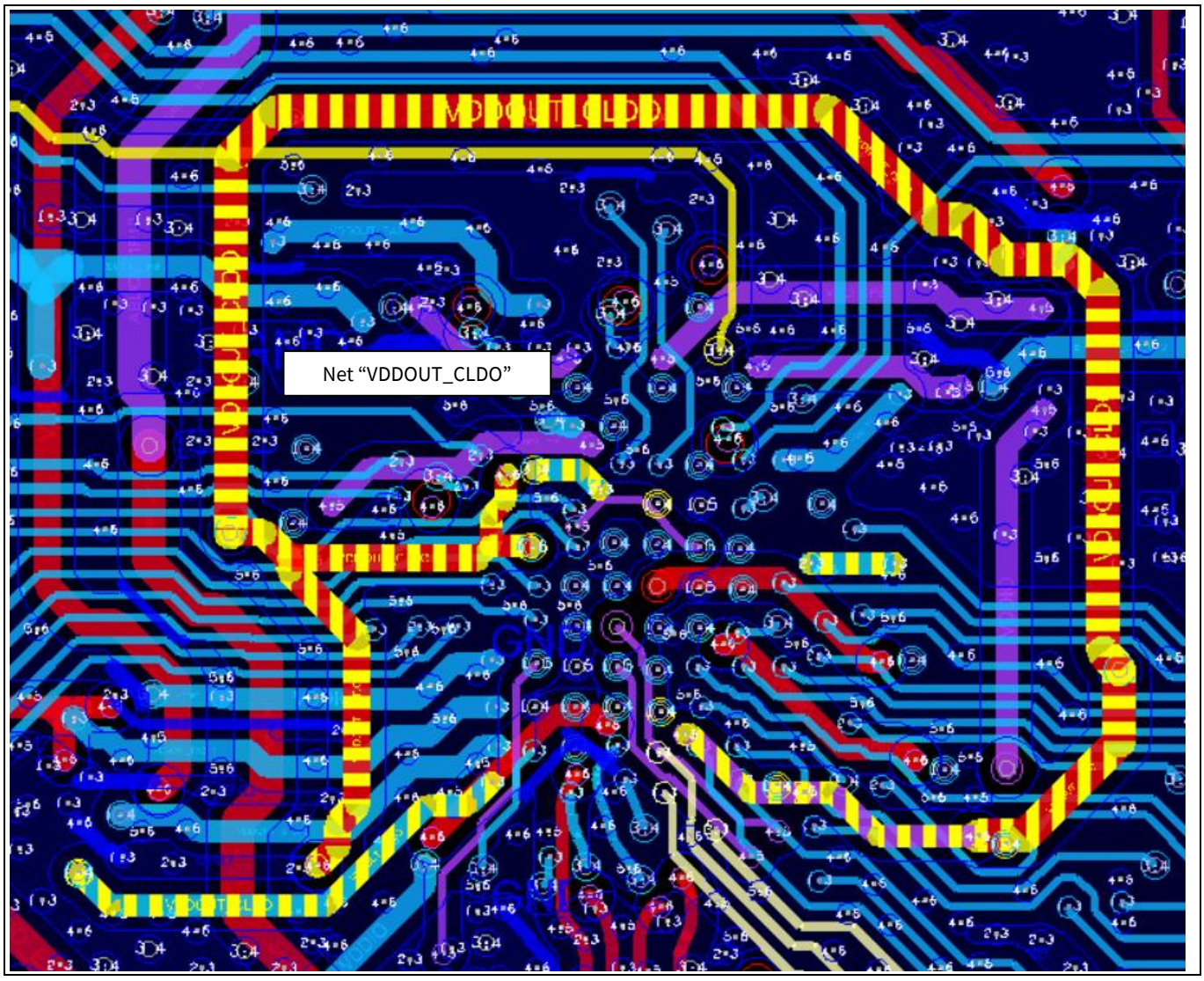


Figure 26 VDDOUT_CLDO routing (layers 3, 5, and 6)

4.8 Routing the crystal traces

When routing the crystal (XTAL) traces, consider the following:

- Ensure that XTAL_IN and XTAL_OUT traces do not overlap.
- Isolate ground as XTAL ground at layer 1 and layer 2, and connect XTAL ground directly to XTAL_GND pins.
- Ensure that the XTAL ground plane has direct vias to the reference ground plane.

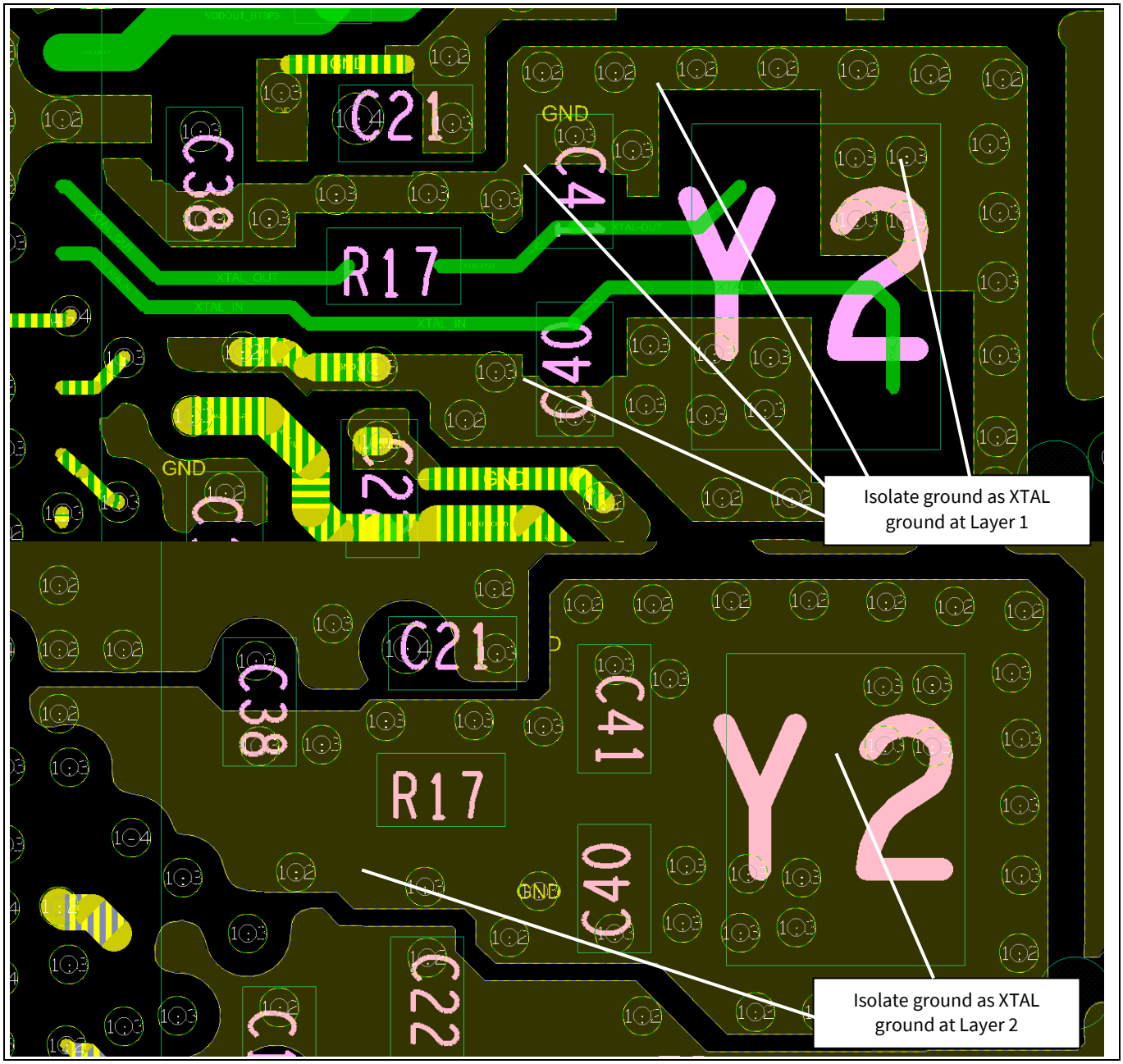


Figure 27 XTAL ground

4.9 Routing the RF traces

When routing the RF traces, consider the following:

- Keep the RF traces on the component layer to avoid using vias.
- Add ground stitches around the RF traces.
- Make sure that adequate isolation and spacing are provided for each RF trace.
- Make sure that the RF traces and stack-up are properly designed so that RF trace impedance is controlled to 50 Ω.
- Make sure that the front-end matching components are placed close to the chip pin.

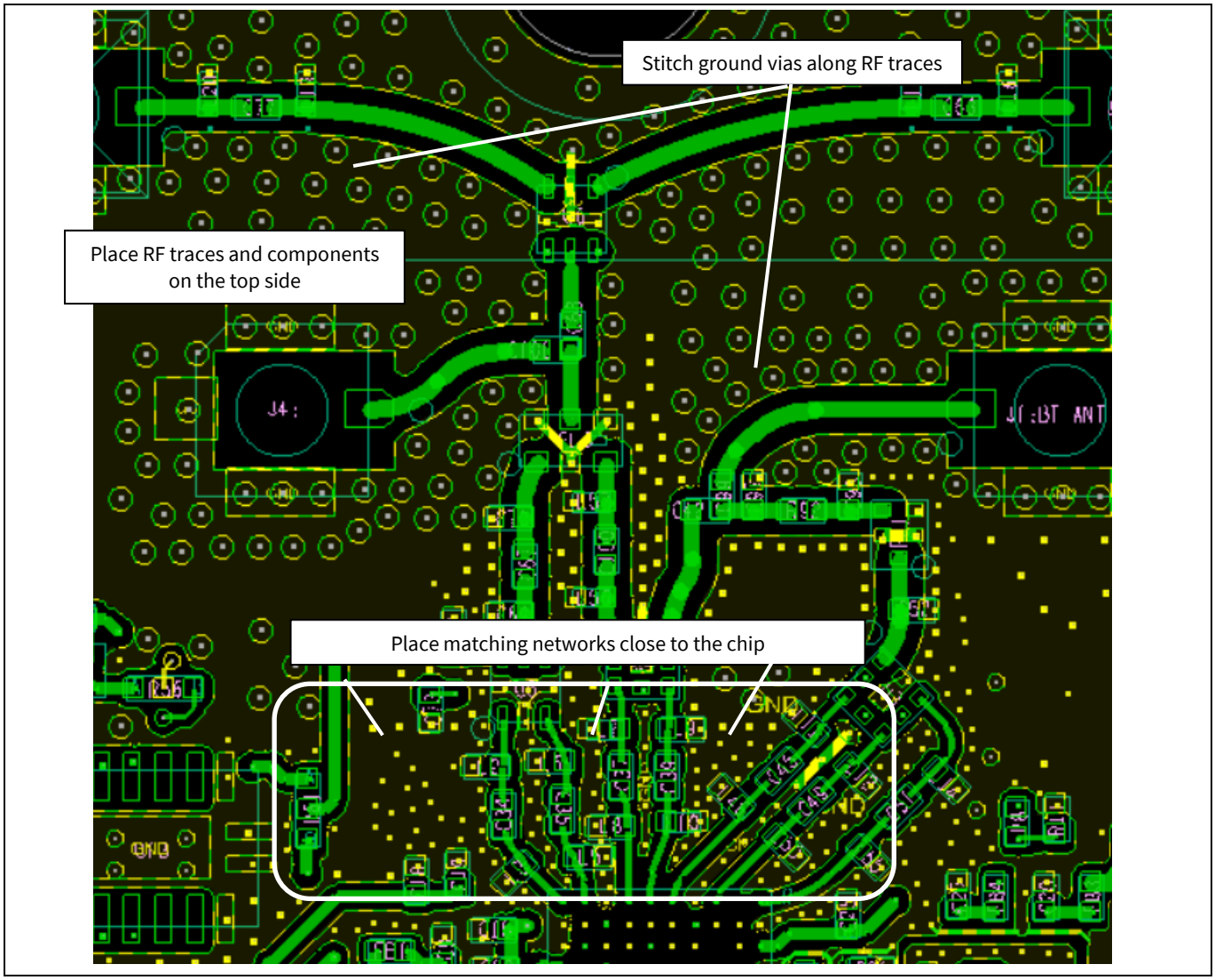


Figure 28 RF component placement and RF trace routing

4.10 RF front end area

Important: DO NOT route any traces in the RF front end area on layers 2 and 3.

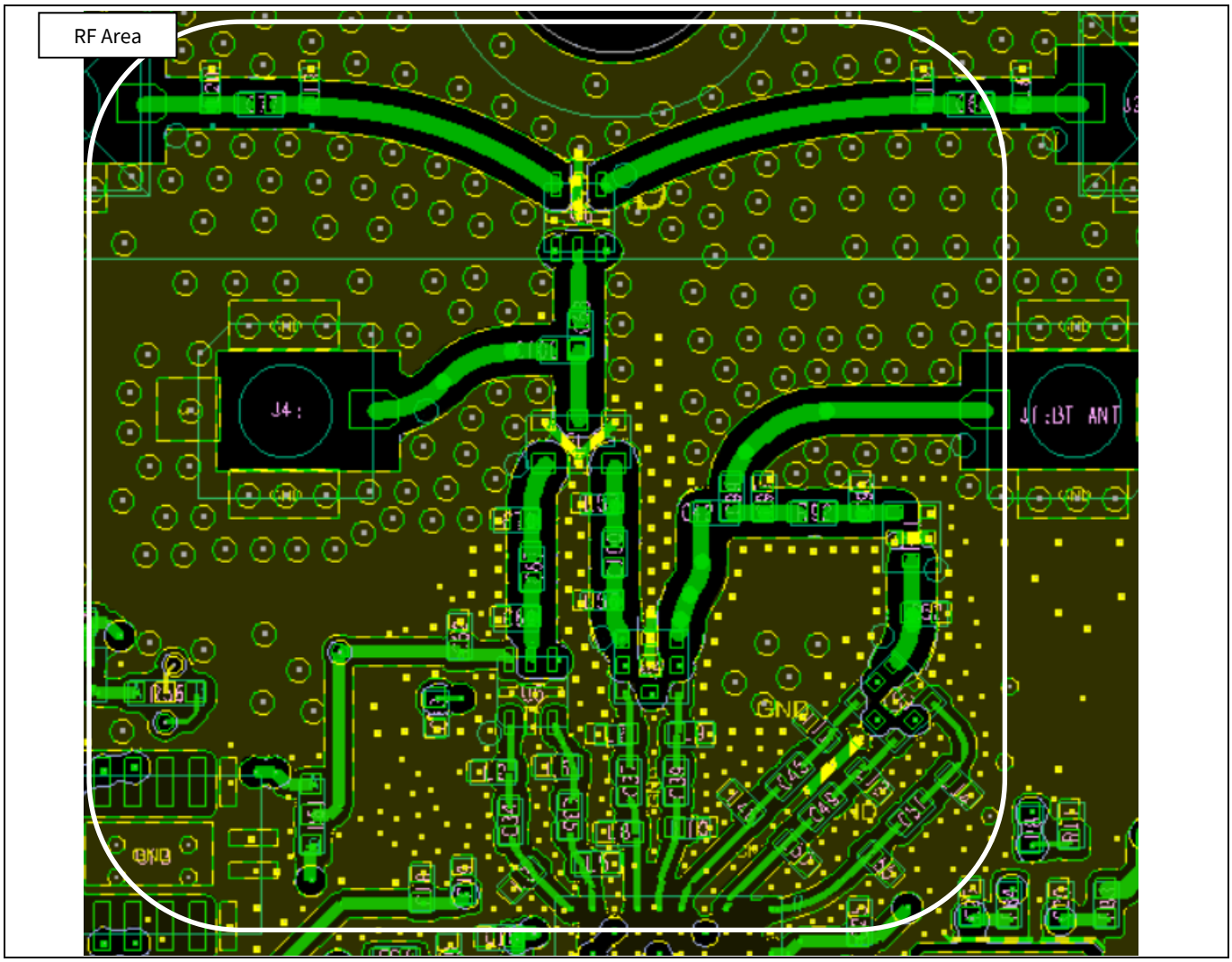


Figure 29 RF area routing keepout

4.11 Strapping and Routing of the SDIO interfaces

SDIO_DATA_2 signal can be used as a strapping option for selecting between SDIO and gSPI modes as below. Default mode is SDIO.

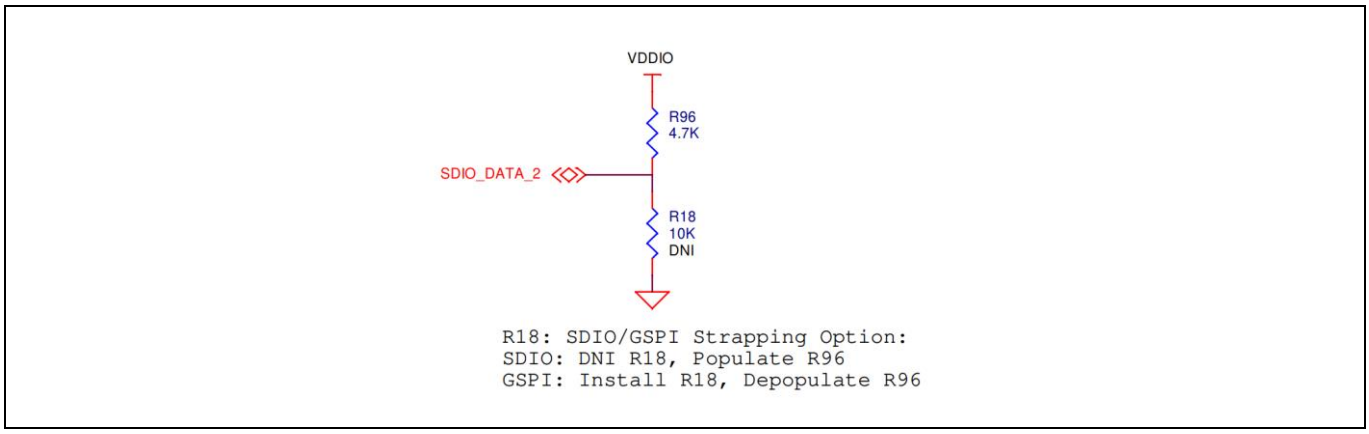


Figure 30 Strapping using the SDIO_DATA_2 signal

Consider the following when designing the layout of the SDIO interfaces:

- Do not route digital traces directly below the PMU, RF, and crystal areas.
- Isolate the SDIO bus, particularly the SDIO clock, from noisy signals. The SDIO clock trace should also be isolated from the other SDIO traces if possible.
- Maintain the impedance of SDIO traces as 50 Ω, single-end.
- Make sure that the length of SDIO data traces matches the SDIO clock trace within ±100 mils.

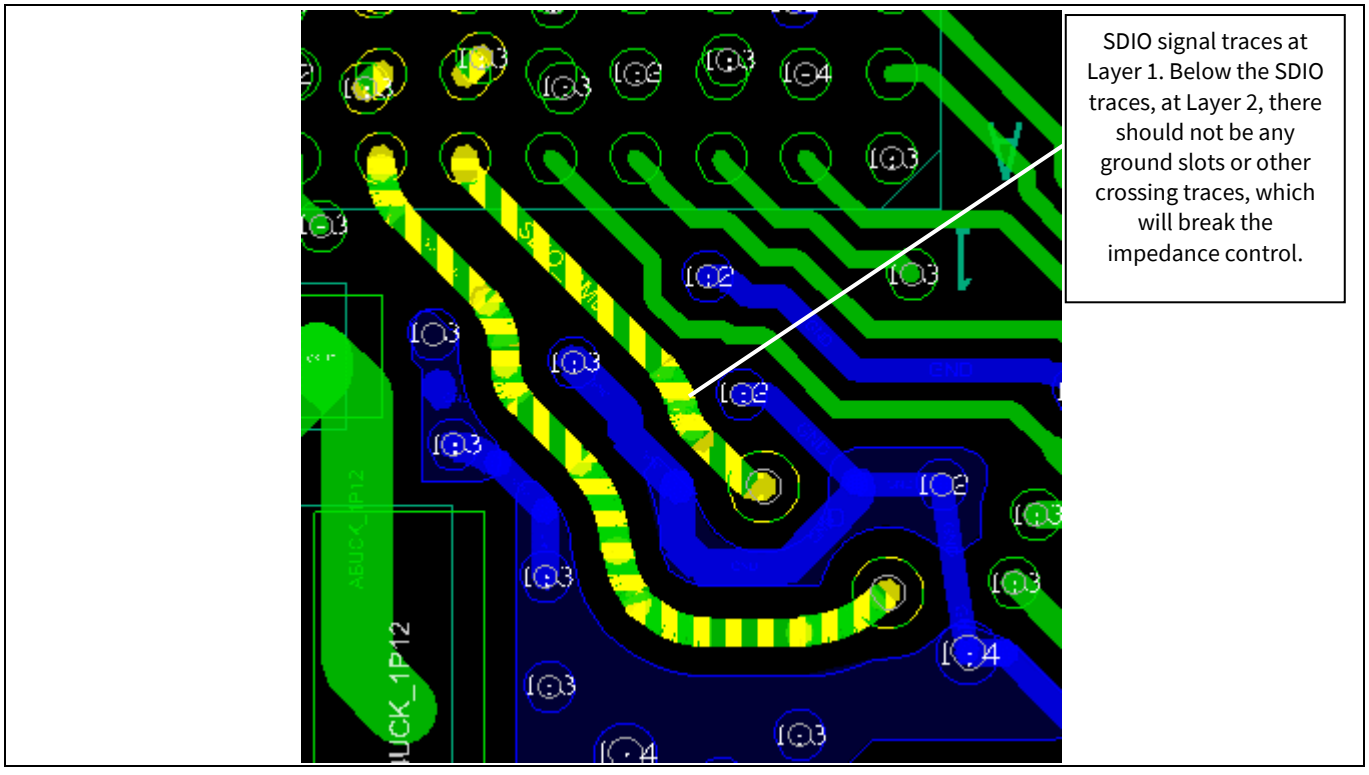


Figure 31 Layout of SDIO traces (top layer)

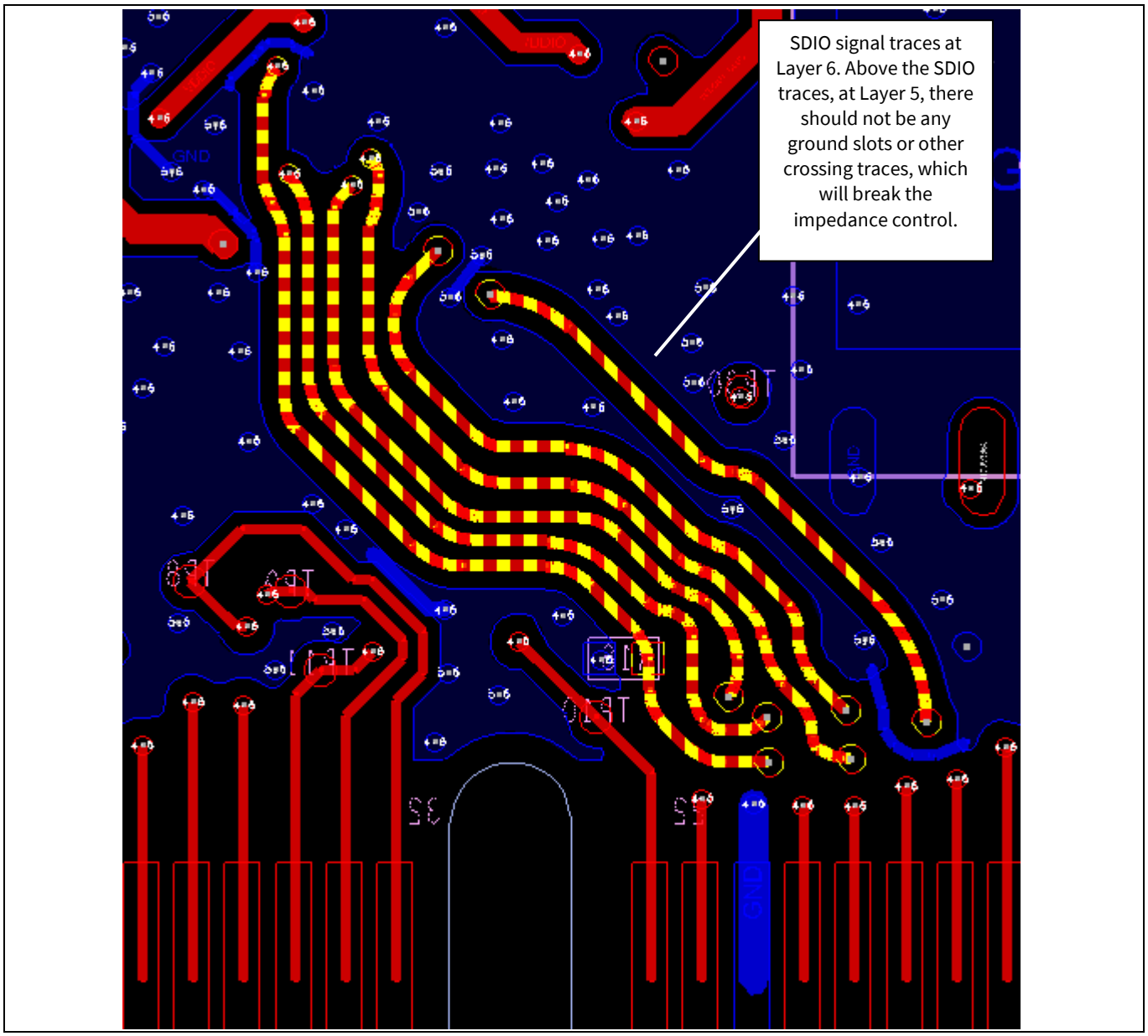


Figure 32 Layout of SDIO traces (bottom layer)

4.12 Routing the SMIF signals

SMIF signals can be used to connect external serial devices such as Serial FLASH and PSRAM. The SMIF DATA and CLK signals can be shared between these serial devices; whereas SMIF_SPHB_CS0_N and SMIF_SPHB_CS1_N can be used to select the serial FLASH and PSRAM individually.

- Ensure to connect the CLK signal of the CYW5591x to serial FLASH and PSRAM via a 33 ohms resistor.
- Trace length matching and control is required for SMIF signals connecting to serial FLASH and PSRAM separately during layout routing.
- Ensure to keep minimum trace length for CLK signals shared between serial FLASH and PSRAM.

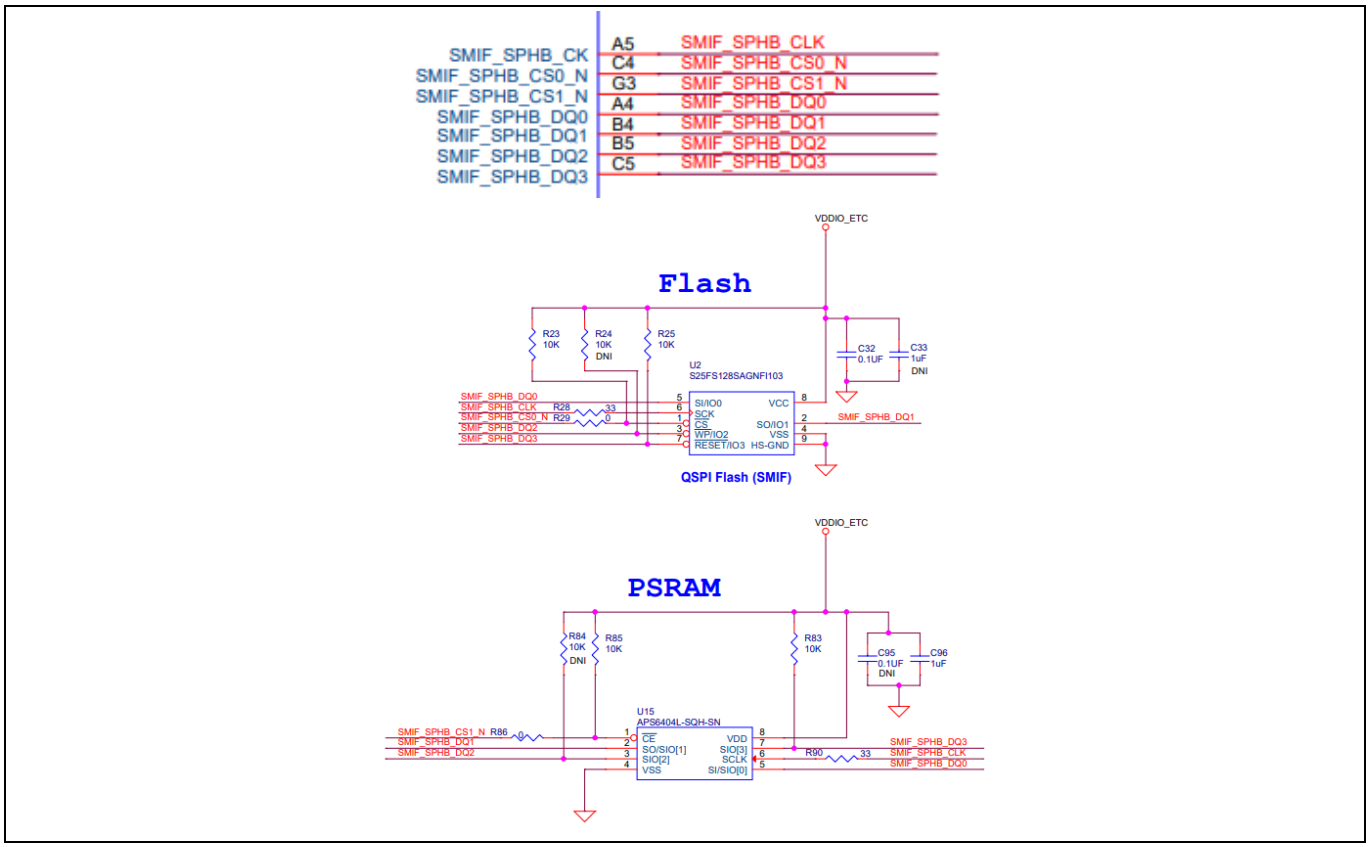


Figure 33 SMIF signals connections for serial FLASH and PSRAM

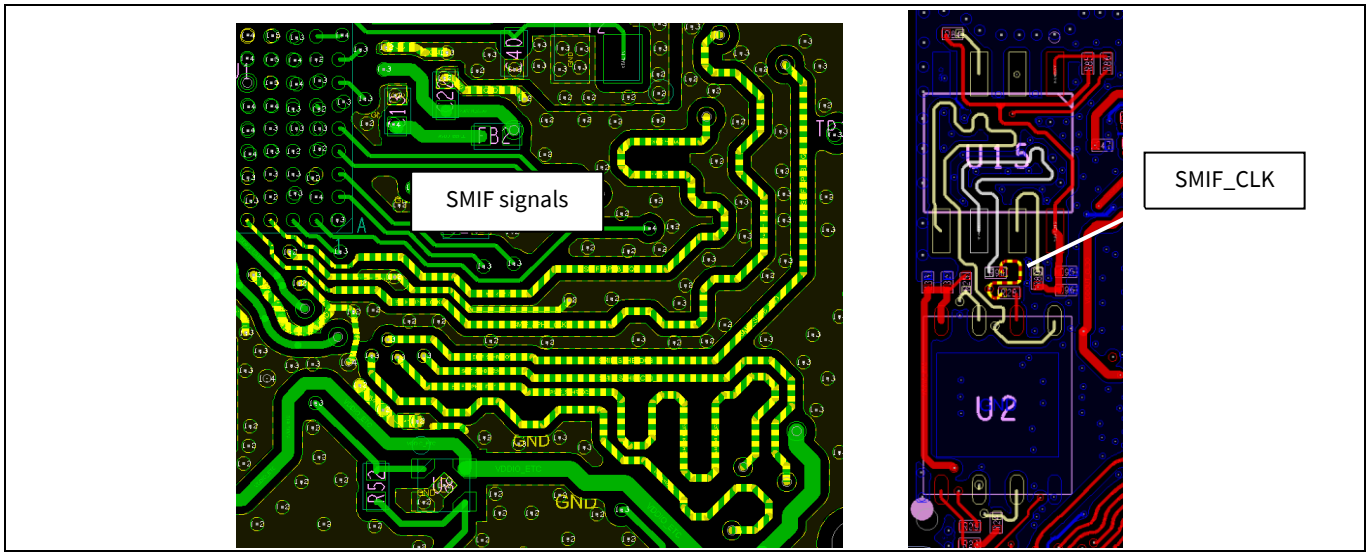


Figure 34 SMIF signals routing for serial FLASH and PSRAM

Routing guidelines

Consider the following when designing the layout of the SMIF signals for PSRAM:

- The clearance of two adjacent signals is 2 times of trace width at least
- Keep the same spacing between ref plane and signal trace in 2 layer PCB design.
- Keep traces as straight as possible.
- Avoid big turn and long serpentine routing when need to tune trace length.
- Avoid routing traces with 90° angle corners.
- Avoid signal trace no reference plane, crossing split or void plane in microstrip line design.
- Signal traces always have continuous reference plane.
- Add reference via around signal via when signal changes to another layer.
- De-capacitor must be placed close to VDD pin.
- GND via is nearby VSS pin.
- De-capacitors would be required with 4.7uF and 0.1uF at least.
- Place the lowest valued capacitor as close as possible to the power pin of the PSRAM.
- Keep power trace width wide.
- Keep power trace lengths short from the package pads to the vias.
- Avoid sharing the same via with two or more decoupling capacitors.
- Avoid Power and GND trace too thin and long.
- De-capacitor must be placed on the current flow path.

Routing guidelines

For reference, the table below provides serial FLASH suggestions that can be supported in the design:

Table 7 Serial FLASH suggestions supported in the design

Vendor	Part Number	Size	Voltage	Package
GigaDevice	GD25LE80E	8 Mb	1.65 to 3.6 V	3x2 mm USON8
GigaDevice	GD25LE16E	16 Mb	1.65 V to 3.6 V	3x2 mm USON8
GigaDevice	GD25LE32E	32 Mb	1.65 V to 3.6 V	6x5 mm WSON8
GigaDevice	GD25LE64E	64 Mb	1.65 V to 3.6 V	6x5 mm WSON8
GigaDevice	GD25LE128E	128 Mb	1.6 V to 2.0 V	6x5 mm WSON8
Infineon	S25FS064S	64 Mb	1.8 V	6x5 mm LGA
Infineon	S25FS128S	128 Mb	1.8 V	6x5 mm WSON
Infineon	ULP RRAM	8 to 128 Mb	1.8 V	USON
ISSI	IS25WP080	8 Mb	1.8 V	6x5 WSON/2x3USON8
ISSI	IS25WP016D	16 Mb	1.8 V	6x5 WSON
ISSI	IS25WP032D	32 Mb	1.8 V	6x5 WSON
ISSI	IS25WP64D	64 Mb	1.8 V	6x5 WSON
ISSI	IS25WP128D	128 Mb	1.8 V	6x5 WSON
Macronix	MX25U8035F	8 Mb	1.8 V	2x3 USON8
Macronix	MX25U1632F	16 Mb	1.8 V	2x3 USON8
Macronix	MX25U3232F	32 Mb	1.8 V	6x5 WSON
Macronix	MX25U6432F	64 Mb	1.8 V	6x5 WSON
Macronix	MX25U12832F	128 Mb	1.8 V	6x5 WSON
Winbond	W25Q80EW	8 Mb	1.8 V	6x5 WSON
Winbond	W25Q16FW	16 Mb	1.8 V	6x5 WSON
Winbond	W25Q32FW	32 Mb	1.8 V	6x5 WSON
Winbond	W25Q64FW	64 Mb	1.8 V	6x5 WSON
Winbond	W25Q128FW	128 Mb	1.8 V	6x5 WSON

Table 8 PSRAM suggestions supported in the design

PSRAM parts	TSUI	THOI	TCK-Q	THO	TCSS	TCSH
CYW55913 SDR mode SMIF: QSPI timing specs	2	3	7	0	5	5
IPUS IPS6404LSQ	2	2	6	1.5	3	3
APS6404L-SQH (supports half sleep)	2	2	5.5	1.5	2.5	3
APS6404L-SQN	2	2	5.5	1.5	2.5	3

Note: Any QSPI 1.8 V PSRAM that meets the timing requirements of the CYW559xx QSPI interface can be utilized.

References

Contact Infineon support to access technical documentation and software.

[1] Infineon Technologies AG: Datasheet – *AIROC™ Wi-Fi & Bluetooth® LE Connected MCU Low-power, 1 x 1, tri-band, IEEE 802.11ax, Wi-Fi 6/6E, Bluetooth® Low Energy 5.4*

[2] Infineon Technologies AG: *CYW955913SDCM2WLIPA design pack;* Available online

Revision history

Document revision	Date	Description of changes
**	2024-10-27	Initial release

Trademarks

All referenced product or service names and trademarks are the property of their respective owners.

The Bluetooth® word mark and logos are registered trademarks owned by Bluetooth SIG, Inc., and any use of such marks by Infineon is under license.

Edition 2024-10-27

Published by

Infineon Technologies AG

81726 Munich, Germany

© 2024 Infineon Technologies AG. All Rights Reserved.

Do you have a question about this document?

Email: erratum@infineon.com

Document reference

002-38930 Rev. **

Important notice

The information contained in this application note is given as a hint for the implementation of the product only and shall in no event be regarded as a description or warranty of a certain functionality, condition or quality of the product. Before implementation of the product, the recipient of this application note must verify any function and other technical information given herein in the real application. Infineon Technologies hereby disclaims any and all warranties and liabilities of any kind (including without limitation warranties of non-infringement of intellectual property rights of any third party) with respect to any and all information given in this application note.

The data contained in this document is exclusively intended for technically trained staff. It is the responsibility of customer's technical departments to evaluate the suitability of the product for the intended application and the completeness of the product information given in this document with respect to such application.

Warnings

Due to technical requirements products may contain dangerous substances. For information on the types in question please contact your nearest Infineon Technologies office.

Except as otherwise explicitly approved by Infineon Technologies in a written document signed by authorized representatives of Infineon Technologies, Infineon Technologies' products may not be used in any applications where a failure of the product or any consequences of the use thereof can reasonably be expected to result in personal injury.